

Draft Notes

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0 Introduction and Definitions

GNEREA TODO: migliorare notazione

Here are the operators:

$$\mathbb{T} : \mu(dx, da) \rightarrow \nu(dx') \quad (1)$$

$$\mathbb{P}_\pi : \nu(dx) \rightarrow \pi(da|x) \nu(dx) \quad (2)$$

Let's define:

$$M_\pi = \mathbb{T}\mathbb{P}_\pi \quad (3)$$

0.1 Discounted Occupancy Measure

Let's define the discounted occupancy measure as:

$$M(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{E}_{\substack{s_0 \sim \nu_0(\cdot), \\ a_t \sim \pi(\cdot|s_t), \\ s_{t+1} \sim \mathcal{T}(\cdot|s_t, a_t)}} [\mathbb{1}(s_t = s)] \quad (4)$$

where $(1 - \gamma)$ serves as a normalization term, π is the policy and $\mathcal{T}$ is the transition probability. Now, let's reformulate it using the operators mentioned above:

$$\mathbb{E}_{\substack{s_0 \sim \nu_0(\cdot), \\ a_t \sim \pi(\cdot|s_t), \\ s_{t+1} \sim \mathcal{T}(\cdot|s_t, a_t)}} [\mathbb{1}(s_t = s)] = \sum_{s_0 \in \mathcal{S}} \nu_0(s_0) \sum_{a_0 \in \mathcal{A}} \pi(a_0|s_0) \sum_{s_1 \in \mathcal{S}} \mathcal{T}(s_1|s_0, a_0) \dots \sum_{a_{t-1} \in \mathcal{A}} \pi(a_{t-1}|s_{t-1}) \mathcal{T}(s|s_{t-1}, a_{t-1}) \quad (5)$$

$$= [(\mathbb{T}\mathbb{P})^t \nu_0](s) \quad (6)$$

Then, we can plug this result in the discounted occupancy measure above:

$$M(s) = (1 - \gamma) \sum_{t=0}^{\infty} [(\gamma \mathbb{T}\mathbb{P})^t \nu_0](s) = (1 - \gamma) [(\text{Id} - \gamma M_\pi)^{-1} \nu_0](s) \quad (7)$$

1 Distribution Matching

We can formulate the policy learning objective in an unsupervised manner (i.e., agnostic to a downstream task), as:

$$\nu^* = (1 - \gamma)(\text{Id} - \gamma M_{\pi^*})^{-1} \nu_0 \quad (8)$$

where ν_0 is the starting state distribution and ν^* is the state distribution evolving the initial distributions following the operators $\mathbb{T}$ and $\mathbb{P}_{\pi^*}$. Setting the desired state distribution ν^* as a *uniform* distribution, we could define a minimization optimization problem over the policy as:

$$\pi = \underset{\pi}{\operatorname{argmin}} D(\nu_\pi, \nu^*) = \underset{\pi}{\operatorname{argmin}} D((1 - \gamma)(\text{Id} - \gamma M_\pi)^{-1} \nu_0, \nu^*) \quad (9)$$

MPR: Stiamo lavorando con $(\text{Id} - \gamma M_{\pi^*})^{-1} \nu_0$ discounted state distribution, una discounted "uniform" distribution è uguale ad una uniform distribution?

where D is a distance measure between distributions. In this case, we end up in the category of data-based unsupervised RL algorithms defined in the URLB: Benchmarking ..., but with a new - to the best of our knowledge - operator approach. On the other hand, we could measure the state visited distribution of an expert policy and set ν^* as the one obtained by the policy expert π^* .

$$\nu^* = (\text{Id} - \gamma \text{TP}_{\pi^*})^{-1} \nu_0 \quad (10)$$

$$\nu_0 = (\text{Id} - \gamma \text{TP}_{\hat{\pi}}) \hat{\nu} \quad (11)$$

Setting that our target distribution $\hat{\nu} = \nu^*$, we obtain that: missing $(1 - \gamma)$

$$\nu_0 = (\text{Id} - \gamma \text{TP}_{\hat{\pi}})(\text{Id} - \gamma \text{TP}_{\pi^*})^{-1} \nu_0 \quad (12)$$

$$\nu_0 = (\text{Id} - \gamma \text{TP}_{\hat{\pi}} \pm \gamma \text{TP}_{\pi^*})(\text{Id} - \gamma \text{TP}_{\pi^*})^{-1} \nu_0 \quad (13)$$

$$\nu_0 = \nu_0 + \gamma \text{T}(\text{P}_{\pi^*} - \text{P}_{\hat{\pi}})(\text{Id} - \gamma \text{TP}_{\pi^*})^{-1} \nu_0 \quad (14)$$

$$0 = \gamma \text{T}(\text{P}_{\pi^*} - \text{P}_{\hat{\pi}})(\text{Id} - \gamma \text{TP}_{\pi^*})^{-1} \nu_0 \quad (15)$$

given that T is a non-null operator, that $\gamma \in (0, 1)$ and ν_0 is a non-null state distribution, the solution of the above equation is that the target operator matches the expert policy operator as $\text{P}_{\pi^*} = \text{P}_{\hat{\pi}}$. Here, we see that we end up in an imitation-learning-like approach. Even if this operator formulation looks new and innovative, the matching distribution goal is not (I think).

MPPr: Not clear here what happens if two or more policies have the same occupancy

1.1 Deriving the Policy

Given a target distribution ν^* and a known transition matrix T , missing $(1 - \gamma)$

$$\nu_0 = (\text{Id} - \gamma \text{TP}_{\hat{\pi}}) \nu^* \quad (16)$$

$$\nu^* - \nu_0 = \gamma \text{TP}_{\hat{\pi}} \nu^* \quad (17)$$

$$\text{P}_{\hat{\pi}} = \text{T}^\dagger \frac{\nu^* - \nu_0}{\gamma} \nu^{*\dagger} \quad (18)$$

we know that the pseudo-inverse $\nu^{*\dagger}$ and $\text{T}^\dagger$ brings to sub-stochastic $\text{P}_{\hat{\pi}}$, which breaks the definition of the policy operator.

Then, we can derive the policy using **policy mirror descent** with the following formulation:

$$\text{P}_{\pi_{t+1}} = \underset{\text{P}_\pi}{\text{argmin}} \langle \nabla f(\text{P}_{\pi_t}), \text{P}_\pi - \text{P}_{\pi_t} \rangle + \frac{1}{\eta} D_{\text{KL}}(\text{P}_\pi; \text{P}_{\pi_t}) \quad (19)$$

In this case, the update rule can be computed in closed-form solution as:

$$\text{P}_{\pi_{t+1}}^i = \frac{\text{P}_{\pi_t}^i \exp(-\eta_k \nabla_i f(\text{P}_{\pi_t}))}{\sum_{j=1}^n \text{P}_{\pi_t}^j \exp(-\eta_k \nabla_j f(\text{P}_{\pi_t}))} \quad (20)$$

where f can be set as a distance between distributions $D(\mathcal{P}; \mathcal{Q})$, where $\mathcal{P}$ and $\mathcal{Q}$ are in this case generic.

1.1.1 Kullback-Leibler

Forward KL. missing $(1 - \gamma)$ In this case, we set $f(\text{P}_{\pi_t}) = D_{\text{KL}}\left(\nu_0; \frac{\text{Id} - \gamma \text{TP}_{\pi_t}}{1 - \gamma} \nu^*\right)$ and we want to optimize the following:

$$\text{P}_{\pi_{t+1}} = \underset{\text{P}_\pi}{\text{argmin}} \langle \nabla_{\text{P}_\pi} D_{\text{KL}}\left(\nu_0; \frac{\text{Id} - \gamma \text{TP}_{\pi_t}}{1 - \gamma} \nu^*\right), \text{P}_\pi - \text{P}_{\pi_t} \rangle + \frac{1}{\eta} D_{\text{KL}}(\text{P}_\pi; \text{P}_{\pi_t}) \quad (21)$$

The gradient could be computed in closed form as follows. Let's consider a $\mathcal{X}$ and $\mathcal{A}$ discrete.

$$D_{\text{KL}}(\nu_0; (\text{Id} - \gamma \text{TP}_{\pi_t})\nu^*) = \sum_{x \in \mathcal{X}} \nu_0(x) \log \frac{\nu_0(x)}{\frac{\text{Id} - \gamma \text{TP}_{\pi_t}}{1 - \gamma} \nu^*} \quad (22)$$

$$= \sum_{x \in \mathcal{X}} \nu_0(x) \log \frac{\nu_0(x)}{Q(x)} \quad (23)$$

$$D_{\text{KL}}(\nu_0; Q(x)) = \sum_{x \in \mathcal{X}} \nu_0(x) \log \frac{\nu_0(x)}{Q(x)} \quad (24)$$

$$\frac{\delta D_{\text{KL}}(\nu_0; Q(x))}{\delta Q(x)} = -\frac{\nu_0(x)}{Q(x)} = -r(x) \quad (25)$$

$$\delta Q(x) = -\gamma \frac{\text{T} \delta \text{P}_{\pi} \nu^*}{1 - \gamma} \quad (26)$$

$$\delta f = \text{Tr}((\nabla_{\text{P}_{\pi}} f)^{\top} \delta \text{P}_{\pi}) \quad (27)$$

$$\delta D_{\text{KL}}(\nu_0; Q(x)) = -r^{\top} dQ_{\text{P}_{\pi}} = r^{\top} \gamma \frac{\text{T} \delta \text{P}_{\pi} \nu^*}{1 - \gamma} \quad (28)$$

$$= \frac{\text{Tr}(r^{\top} \gamma \text{T} \delta \text{P}_{\pi} \nu^*)}{1 - \gamma} = \frac{\text{Tr}(r^{\top} \gamma \text{T} \delta \text{P}_{\pi} \nu^*)}{1 - \gamma} = \frac{\text{Tr}(\nu^* r^{\top} \gamma \text{T} \delta \text{P}_{\pi})}{1 - \gamma} \quad (29)$$

$$\nabla_{\text{P}_{\pi}} D_{\text{KL}}(\nu_0; (\text{Id} - \gamma \text{TP}_{\pi_t})) = \frac{\gamma \text{T}^{\top} r(\nu^*)^{\top}}{1 - \gamma} \quad (30)$$

In Eq. 22, the KL definition. In Eq. 24, define the denominator as Q . In Eq 25, the forward KL derivative plus the r definition as that ratio. Then, in Eq. 26, the derivative is linear with respect to that linear operator. Matrix differential calculus uses Eq. 38 to identify the gradient under the Frobenius inner product. In eq 29, any scalar equals its trace, so $r^{\top} T \delta_{\Pi} \nu^* = \text{Tr}(r^{\top} T \delta_{\Pi} \nu^*)$. Using cyclicity of trace, $\text{Tr}(r^{\top} T \delta_{\Pi} \nu^*) = \text{Tr}((\nu^*)^{\top} r^{\top} T \delta_{\Pi})$.

Reverse KL. In this case, we set $f(\text{P}_{\pi_t}) = D_{\text{KL}}((1 - \gamma)(\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} \nu_0; \nu^*)$ and we want to optimize the following:

$$\text{P}_{\pi_{t+1}} = \underset{\text{P}_{\pi}}{\text{argmin}} \langle \nabla_{\text{P}_{\pi}} D_{\text{KL}}((1 - \gamma)(\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} \nu_0; \nu^*), \text{P}_{\pi} - \text{P}_{\pi_t} \rangle + \frac{1}{\eta} D_{\text{KL}}(\text{P}_{\pi}; \text{P}_{\pi_t}) \quad (31)$$

Per calcolare il gradiente:

$$D_{\text{KL}}((1 - \gamma)(\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} \nu_0; \nu^*) = D_{\text{KL}}(\nu_{\pi}; \nu^*) \quad (32)$$

$$\nabla_{\nu_{\pi}} D_{\text{KL}}(\nu_{\pi}; \nu^*) = 1 + \log \frac{\nu_{\pi}(x)}{\nu^*(x)} \quad (33)$$

$$\frac{\delta \nu_{\pi}(x)}{\delta \text{P}_{\pi}} = \frac{\delta(1 - \gamma)(\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} \nu_0}{\delta \text{P}_{\pi}} = \frac{(1 - \gamma) \delta A^{-1} \nu_0}{\delta \text{P}_{\pi}} \quad (34)$$

$$\delta A^{-1} = -A^{-1} \delta A A^{-1} = (\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} (\gamma \text{T} \delta \text{P}_{\pi_t}) (\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} \quad (35)$$

$$\frac{\delta \nu_{\pi}(x)}{\delta \text{P}_{\pi}} = \frac{(1 - \gamma) \delta A^{-1} \nu_0}{\delta \text{P}_{\pi}} = \frac{(1 - \gamma)(\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} (\gamma \text{T} \delta \text{P}_{\pi_t}) (\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} \nu_0}{\delta \text{P}_{\pi}} \quad (36)$$

$$\delta D_{\text{KL}}(\nu_{\pi}; \nu^*) = \left(1 + \log \frac{\nu_{\pi}(x)}{\nu^*(x)} \right)^{\top} (1 - \gamma)(\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} (\gamma \text{T} \delta \text{P}_{\pi_t}) (\text{Id} - \gamma \text{TP}_{\pi_t})^{-1} \nu_0 \quad (37)$$

$$\delta f = \text{Tr}((\nabla_{\mathbf{P}_\pi} f)^\top \delta \mathbf{P}_\pi) \quad (38)$$

$$\delta D_{\text{KL}}(\nu_\pi; \nu^*) = (1 - \gamma) \text{Tr} \left[\left(1 + \log \frac{\nu_\pi(x)}{\nu^*(x)} \right)^\top (\text{Id} - \gamma \mathbf{TP}_{\pi_t})^{-1} (\gamma \mathbf{T} \delta \mathbf{P}_\pi) (\text{Id} - \gamma \mathbf{TP}_{\pi_t})^{-1} \nu_0 \right] \quad (39)$$

$$= (1 - \gamma) \text{Tr} \left[(\text{Id} - \gamma \mathbf{TP}_{\pi_t})^{-1} \nu_0 \left(1 + \log \frac{\nu_\pi(x)}{\nu^*(x)} \right)^\top (\text{Id} - \gamma \mathbf{TP}_{\pi_t})^{-1} (\gamma \mathbf{T} \delta \mathbf{P}_\pi) \right] \quad (40)$$

$$= (1 - \gamma) \text{Tr} \left[\left((\gamma \mathbf{T})^\top [(\text{Id} - \gamma \mathbf{TP}_{\pi_t})^{-1}]^\top \left(1 + \log \frac{\nu_\pi(x)}{\nu^*(x)} \right) \nu_0^\top [(\text{Id} - \gamma \mathbf{TP}_{\pi_t})^{-1}]^\top \right)^\top \delta \mathbf{P}_\pi \right] \quad (41)$$

$$\nabla_{\mathbf{P}_\pi} D_{\text{KL}}((1 - \gamma)(\text{Id} - \gamma \mathbf{TP}_{\pi_t})^{-1} \nu_0; \nu^*) = (1 - \gamma) \gamma \mathbf{T}^\top [(\text{Id} - \gamma \mathbf{TP}_{\pi_t})^{-1}]^\top \left(1 + \log \frac{\nu_\pi(x)}{\nu^*(x)} \right) \nu_0^\top [(\text{Id} - \gamma \mathbf{TP}_{\pi_t})^{-1}]^\top \quad (42)$$

1.2 The need for Maximum-Mean Discrepancy when working with feature embeddings

We consider a (continuous) state space $\mathcal{X}$ and a discrete action space $\mathcal{A}$. We define the embedding functions

$$\phi : \mathcal{X} \rightarrow \mathcal{Z}_x \quad (43)$$

$$\psi : \mathcal{X} \times \mathcal{A} \rightarrow \mathcal{Z}_{x \times a} \quad (44)$$

which map states and state-action pairs, respectively, into their corresponding latent spaces. Since the action space $\mathcal{A}$ is discrete, the joint embedding can be written as $\psi(x, a) = \phi(x) \otimes e_a$, where e_a denotes the one-hot encoding associated with action a .

In the tabular setting, the state embedding function can be defined trivially using one-hot encodings for both states and actions. Moreover, the target distribution is uniform over the state space, i.e., $\nu^*(\cdot) = \frac{1}{|\mathcal{X}|}$, $\forall x \in \mathcal{X}$. However, defining an analogous target discounted state distribution in continuous state spaces is nontrivial, as one-hot encodings are no longer applicable. In the tabular case, each element of the vector $\nu \in \mathbb{R}^{|\mathcal{X}|}$ corresponds uniquely to a state, which allows to directly specify the desired discounted state occupancy values.

Due to the difficulty of specifying a target distribution in the latent space and, consequently, computing the Kullback–Leibler divergence, we instead optimize the maximum mean discrepancy defined as $\|\nu(\phi(x)) - \nu^*(\phi(x))\|_2^2$. By choosing the target distribution to be the null function and adopting one-hot encodings for the state space, this optimization leads to the circumcenter of the embedding simplex. Figure 1 provides a geometric illustration for a three-state environment, with one-hot encoding.

Although this property holds in the tabular setting (more precisely for one-hot encoding), it remains unclear whether the same behavior persists when using learned feature embeddings in continuous state spaces.

Maximum Mean Discrepancy. Here, we compute the gradient for minimizing the maximum mean discrepancy by setting $f(\mathbf{P}_{\pi_t}) = \|\nu(\phi(x))\|_2^2$. We set $\Phi \in \mathbb{R}^{n \times z_x}$, $\Psi \in \mathbb{R}^{n \times z_{xa}}$ as the datasets containing the state and the state-action pair embeddings. We recall that $\mathbf{T} \in \mathbb{R}^{z_x \times z_{xa}}$ is the minimizer of the following equation:

$$\mathbf{T} = \underset{\mathbf{T}}{\text{argmin}} \frac{1}{n} \sum_{i=1}^n \|\varphi(x'_i) - \mathbf{T} \psi(x_i, a_i)\|_2^2 + \lambda \|\mathbf{T}\|^2 = \Phi^* (\Psi \Psi^* + \lambda \text{Id}_n)^{-1} \Psi = \Phi^T B \Psi \quad (45)$$

The closed-form solution is derived from the dual solution of regularized least-square regression. The discounted state distribution can be computed as:

$$\nu_{\mathbf{P}_\pi}(\phi(x)) = (1 - \gamma) \sum_{t=0}^{\infty} [(\gamma \mathbf{TP})^t \nu_0](x) \quad (46)$$

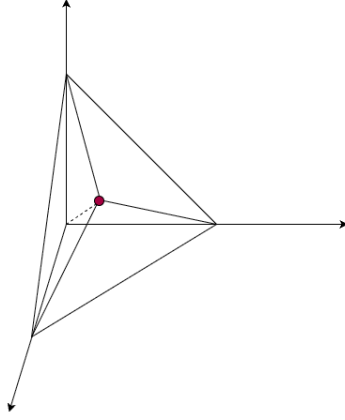


Figure 1: Caption

assuming that $\nu_0 = \Phi^* \alpha$, $\alpha \in \mathbb{R}^n$, we can rewrite the vector of the the discounted state distribution as:

$$\nu_{P_\pi} = (1 - \gamma) \sum_{t=0}^{\infty} (\gamma TP)^t \Phi^* \alpha \quad (47)$$

$$= (1 - \gamma) \sum_{t=0}^{\infty} (\gamma \Phi^* B \Psi P)^t \Phi^* \alpha \quad (48)$$

$$= (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Phi^* (B \Psi P \Phi^*)^t \alpha \quad (49)$$

$$= (1 - \gamma) \Phi^* (\text{Id} - \gamma B \Psi P \Phi^*)^{-1} \alpha \quad (50)$$

$$= (1 - \gamma) \Phi^* (\text{Id} - \gamma BM)^{-1} \alpha \quad (51)$$

Now, we compute the $\nabla_{P_\pi} (\|(1 - \gamma)(\text{Id} - \gamma TP)^{-1} \nu_0\|_2^2)$.

$$\delta \text{MMD} = 2[(1 - \gamma)(\text{Id} - \gamma TP)^{-1} \nu_0]^\top [-(1 - \gamma)(\text{Id} - \gamma TP)^{-1} (-\gamma T \delta P)(\text{Id} - \gamma TP)^{-1} \nu_0] \quad (52)$$

$$= 2\gamma(1 - \gamma)^2 [(\text{Id} - \gamma TP)^{-1} \nu_0]^\top [(\text{Id} - \gamma TP)^{-1} (T \delta P)(\text{Id} - \gamma TP)^{-1} \nu_0] \quad (53)$$

$$(54)$$

$$\delta \text{MMD} = \text{Tr}((\nabla_{P_\pi} f)^\top \delta P_\pi) \quad (55)$$

$$= 2\gamma(1 - \gamma)^2 \text{Tr} \left[[(\text{Id} - \gamma TP)^{-1} \nu_0]^\top [(\text{Id} - \gamma TP)^{-1} (T \delta P)(\text{Id} - \gamma TP)^{-1} \nu_0] \right] \quad (56)$$

$$= 2\gamma(1 - \gamma)^2 \text{Tr} \left[(\text{Id} - \gamma TP)^{-1} \nu_0 [(\text{Id} - \gamma TP)^{-1} \nu_0]^\top [(\text{Id} - \gamma TP)^{-1} (T \delta P)] \right] \quad (57)$$

$$= 2\gamma(1 - \gamma)^2 \text{Tr} \left[(T^\top (\text{Id} - \gamma TP)^{-\top} (\text{Id} - \gamma TP)^{-1} \nu_0 [(\text{Id} - \gamma TP)^{-1} \nu_0]^\top)^\top (\delta P) \right] \quad (58)$$

$$\nabla_{P_\pi} \text{MMD} = 2\gamma(1 - \gamma)^2 T^\top (\text{Id} - \gamma TP)^{-\top} (\text{Id} - \gamma TP)^{-1} \nu_0 [(\text{Id} - \gamma TP)^{-1} \nu_0]^\top \quad (59)$$

$$\nabla_{P_\pi} \text{MMD} = \beta (\Phi^T B \Psi)^\top (\text{Id} - \gamma \Phi^T B \Psi P)^{-\top} (\text{Id} - \gamma \Phi^T B \Psi P)^{-1} \Phi^\top \alpha \quad (60)$$

$$= \beta \Psi^\top B^\top (\text{Id} - \gamma BM)^{-\top} \Phi \Phi^\top (\text{Id} - \gamma BM)^{-1} \alpha \quad (61)$$

MPr: Not clear why we erase the last term to make PMD well-posed

where $\beta = 2\gamma(1 - \gamma)^2$ and $M = \Psi P \Phi^\top$

1.3 Convergence of PMD

For my own convenience let me define °

$$M(P) = (\text{Id} - \gamma TP)^{-1} \quad (62)$$

and the MMD is just $\|M(\mathbf{P})\nu_0\|^2$. I'll also use the fact that

$$\|p\|^2 - \|q\|^2 = \langle p - q, p + q \rangle \quad (63)$$

$$= \|p - q\|^2 + 2 \langle p - q, q \rangle \quad (64)$$

$$= -\|q - p\|^2 + 2 \langle p - q, p \rangle \quad (65)$$

Difference lemma.

$$\begin{aligned} \|M(\mathbf{P}_1)\nu_0\|^2 - \|M(\mathbf{P}_2)\nu_0\|^2 &= \langle (M(\mathbf{P}_1) - M(\mathbf{P}_2))\nu_0, (M(\mathbf{P}_1) + M(\mathbf{P}_2))\nu_0 \rangle \\ &= \gamma \langle M(\mathbf{P}_1)T(\mathbf{P}_1 - \mathbf{P}_2)M(\mathbf{P}_2)\nu_0, (M(\mathbf{P}_1) + M(\mathbf{P}_2))\nu_0 \rangle \\ &= \gamma \langle M(\mathbf{P}_2)\nu_0, (\mathbf{P}_1 - \mathbf{P}_2)^*T^*M(\mathbf{P}_1)^*(M(\mathbf{P}_1) + M(\mathbf{P}_2))\nu_0 \rangle \end{aligned}$$

Directional derivatives.

$$\lim_{h \rightarrow 0} \frac{\|M(\mathbf{P}_1 + h(\mathbf{P}_2 - \mathbf{P}_1))\nu_0\|^2 - \|M(\mathbf{P}_1)\nu_0\|^2}{h} = 2\gamma \langle M(\mathbf{P}_1)T(\mathbf{P}_2 - \mathbf{P}_1)M(\mathbf{P}_1)\nu_0, M(\mathbf{P}_1)\nu_0 \rangle \quad (66)$$

$$= 2\gamma \langle M(\mathbf{P}_1)\nu_0, (\mathbf{P}_2 - \mathbf{P}_1)^*T^*M(\mathbf{P}_1)^*M(\mathbf{P}_1)\nu_0 \rangle \quad (67)$$

Mirror Descent. The PMD iteration at step $k + 1$ is written (pointwise) as

$$\mathbf{P}_{k+1} = \underset{\mathbf{P}}{\operatorname{argmin}} 2\gamma \mathbf{P}^*T^*M(\mathbf{P}_k)^*M(\mathbf{P}_k)\nu_0 + D_{\text{KL}}(\mathbf{P}; \mathbf{P}_k). \quad (68)$$

3-point. I'm jumping to the solution since the proof is exactly the same as in POWR, I just list every step for reference. By the 3-point lemma we have that:

$$2\gamma(\mathbf{P}^* - \mathbf{P}_{k+1}^*)T^*M(\mathbf{P}_k)^*M(\mathbf{P}_k)\nu_0 \geq D_{\text{KL}}(\mathbf{P}; \mathbf{P}_{k+1}) - D_{\text{KL}}(\mathbf{P}; \mathbf{P}_k) \quad (69)$$

$\mathbf{P} = \mathbf{P}_k$ this simplifies to:

$$2\gamma(\mathbf{P}_k^* - \mathbf{P}_{k+1}^*)T^*M(\mathbf{P}_k)^*M(\mathbf{P}_k)\nu_0 \geq 0. \quad (70)$$

To prove a descent condition on the MMD we also need to prove the following

$$2\gamma(\mathbf{P}_k^* - \mathbf{P}_{k+1}^*)T^*M(\mathbf{P}_k)^*M(\mathbf{P}_{k+1})\nu_0 \geq 0. \quad (71)$$

$$\langle M(\mathbf{P}_{k+1})\nu_0, (\mathbf{P}_k^* - \mathbf{P}_{k+1}^*)T^*M(\mathbf{P}_k)^*M(\mathbf{P}_{k+1})\nu_0 \rangle \geq 0. \quad (72)$$

Alternatively,

$$\begin{aligned} \|M(\mathbf{P}_1)\|^2 - \|M(\mathbf{P}_2)\|^2 &= \langle M(\mathbf{P}_1), M(\mathbf{P}_1) \rangle - \langle M(\mathbf{P}_2), M(\mathbf{P}_2) \rangle \\ &= \langle M(\mathbf{P}_1) - M(\mathbf{P}_2), M(\mathbf{P}_1) \rangle + \langle M(\mathbf{P}_2), M(\mathbf{P}_1) - M(\mathbf{P}_2) \rangle \end{aligned}$$

The first term is

$$\begin{aligned} \langle M(\mathbf{P}_1) - M(\mathbf{P}_2), M(\mathbf{P}_1) \rangle &= \langle M(\mathbf{P}_1)(M(\mathbf{P}_2)^{-1} - M(\mathbf{P}_1)^{-1})M(\mathbf{P}_2), M(\mathbf{P}_1) \rangle \\ &= \gamma \langle M(\mathbf{P}_1)T(\mathbf{P}_1 - \mathbf{P}_2)M(\mathbf{P}_2), M(\mathbf{P}_1) \rangle \end{aligned}$$

which we know it's ≥ 0 if $\mathbf{P}_1 = \mathbf{P}_k$ and $\mathbf{P}_2 = \mathbf{P}_{k+1}$.

The second term

$$\begin{aligned} \langle M(\mathbf{P}_2)\nu_0, (M(\mathbf{P}_1) - M(\mathbf{P}_2))\nu_0 \rangle &= \langle M(\mathbf{P}_2)\nu_0, M(\mathbf{P}_2)(M(\mathbf{P}_2)^{-1} - M(\mathbf{P}_1)^{-1})M(\mathbf{P}_1)\nu_0 \rangle \\ &= \langle M(\mathbf{P}_2)T(\mathbf{P}_1 - \mathbf{P}_2)M(\mathbf{P}_1)\nu_0, M(\mathbf{P}_2)\nu_0 \rangle \\ &= \langle M(\mathbf{P}_1)\nu_0, (\mathbf{P}_1 - \mathbf{P}_2)^*T^*M(\mathbf{P}_2)^*M(\mathbf{P}_2)\nu_0 \rangle \end{aligned}$$

When you stay still, then the PMD objective is just:

$$2\gamma P_{k+1}^* T^* M(P_{k+1})^* M(P_{k+1}) \nu_0 \quad (73)$$

when you move back to k then

$$2\gamma P_k^* T^* M(P_{k+1})^* M(P_{k+1}) \nu_0 + D_{KL}(P_k; P_{k+1}) \quad (74)$$

We want

$$2\gamma(P_{k+1}^* - P_k^*) T^* M(P_{k+1})^* M(P_{k+1}) \nu_0 \leq D_{KL}(P_k; P_{k+1}) \quad (75)$$

1.3.1 The Update Rule: Deriving $\Psi^\top c$ in compact form

We consider a dataset $\{(x_i, a_i)\}_{i=1}^n$ with discrete action space $\mathcal{A} = \{1, \dots, A\}$ and a state feature map $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$. Following the joint state-action embedding used in these notes, the feature map on (x, a) is defined as a tensor product between the state feature and the one-hot encoding of the action,

$$\psi(x, a) = \phi(x) \otimes \delta_a \in \mathbb{R}^{dA}, \quad (76)$$

where $\delta_a = e_a \in \mathbb{R}^A$ is the a -th canonical basis vector. This construction is convenient because it yields a block structure in which each action selects a different block of d coordinates while sharing the same state feature $\phi(x)$.

Define the matrices

$$\Phi \in \mathbb{R}^{n \times d}, \quad \Phi_{i,:}^\top = \phi(x_i), \quad (77)$$

and

$$E \in \mathbb{R}^{n \times A}, \quad E_{i,:}^\top = \delta_{a_i}, \quad (78)$$

i.e. E is the matrix collecting one-hot action indicators for each sample (this is the same E introduced in Algorithm 1).

Let $\Psi \in \mathbb{R}^{n \times dA}$ be the joint feature matrix with rows

$$\Psi_{i,:}^\top = \psi(x_i, a_i). \quad (79)$$

Given an arbitrary coefficient vector $c \in \mathbb{R}^n$, the quantity $\Psi^\top c$ expands as

$$\Psi^\top c = \sum_{i=1}^n c_i \psi(x_i, a_i) = \sum_{i=1}^n c_i (\phi(x_i) \otimes \delta_{a_i}). \quad (80)$$

To make the action dependence explicit, group the sum by action $a \in \{1, \dots, A\}$:

$$\Psi^\top c = \sum_{a=1}^A \left(\sum_{i: a_i=a} c_i \phi(x_i) \right) \otimes e_a. \quad (81)$$

This shows that $\Psi^\top c$ is a concatenation of A blocks, each block being a weighted sum of state features over the samples whose action equals a . A convenient way to expose this block structure is to reshape $\Psi^\top c \in \mathbb{R}^{dA}$ into a matrix in $\mathbb{R}^{d \times A}$, where each column corresponds to one action:

$$\text{mat}(\Psi^\top c) = \left[\underbrace{\sum_{i: a_i=1} c_i \phi(x_i)}_{\in \mathbb{R}^d} \quad \underbrace{\sum_{i: a_i=2} c_i \phi(x_i)}_{\in \mathbb{R}^d} \quad \cdots \quad \underbrace{\sum_{i: a_i=A} c_i \phi(x_i)}_{\in \mathbb{R}^d} \right]. \quad (82)$$

Each action column can be written using Φ and the a -th indicator column of E (denoted $E_{:,a}$). Using the elementwise product $\odot$,

$$g_a := \sum_{i: a_i=a} c_i \phi(x_i) = \Phi^\top (c \odot E_{:,a}) = \Phi^\top \text{diag}(c) E_{:,a}. \quad (83)$$

Stacking all columns together yields the fully compact expression

$$\boxed{\text{mat}(\Psi^\top c) = \Phi^\top \text{diag}(c) E = \Phi^\top (c \odot E) \in \mathbb{R}^{d \times A}.} \quad (84)$$

Equivalently, vectorizing back to $\mathbb{R}^{dA}$ gives

$$\Psi^\top c = \text{vec}(\Phi^\top \text{diag}(c) E). \quad (85)$$

In the end, Eq 61 can be rewritten in the form of $\Psi^\top c$. Then, if we want to evaluate the gradient in x' , we can multiply Come posso dire tecnicamente che per valutare il gradiente in x' posso usare il kernel evaluation e fare $\langle \phi(x'), \Phi^\top (c \odot E) \rangle$? Devo assumere qualcosa in RKHS?

Interpretation.. The operator $\Psi^\top$ aggregates sample weights c_i into a set of A separate feature sums—one per action—because the tensor product $\phi(x) \otimes \delta_a$ routes each sample (x_i, a_i) into the coordinates associated with action a_i . This is exactly why, in implementations, one can compute action-wise gradients or statistics by masking with E rather than explicitly building the large matrix $\Psi \in \mathbb{R}^{n \times dA}$.

1.3.2 Algorithm

Algorithm 1 Unsupervised Distribution Matching (U-DiM) TODO

Require: discount factor $\gamma \in (0, 1)$, step size $\eta > 0$, kernel function $k(x, x') = \langle \phi(x), \phi(x') \rangle$ with $\phi : \mathcal{X} \rightarrow \mathcal{Z}_a$, initial weights $C_0 = 0 \in \mathbb{R}^{n \times |\mathcal{A}|}$.

```

1:  $T, \text{PMD}$ : training steps, PMD steps,
2: for  $t = 1$  to  $T$  do
3:   compute  $H_t \in \mathbb{R}^{1 \times n}$  such that  $H_j = k(x_t, x_j)$ 
4:    $\mathbf{a}_t \leftarrow \text{argmaxSOFTMAX}(-\eta H_t C_t) \in \mathbb{R}^{1 \times |\mathcal{A}|}$ 
5:    $\mathbf{x}_{t+1} \sim P(\cdot \mid \mathbf{x}_t, \mathbf{a}_t)$ 
6:    $\mathcal{D} \leftarrow \mathcal{D} \cup (\mathbf{x}_t, \mathbf{a}_t, \mathbf{x}_{t+1})$ 
7:    $\text{UPDATEEMBEDDERS}(\mathcal{D}, \sigma_t)$ 
8:   if  $t \bmod \text{update\_every}$  then
9:      $\text{UPDATEACTOR}(\mathcal{D}, \sigma_t)$ 
10:  end if
11: end for
12: procedure  $\text{UPDATEEMBEDDERS}(\mathcal{D}, \sigma)$ 
13:   /* Contrastive Learning orr booo?? */
14: end procedure
15: procedure  $\text{UPDATEACTOR}(\mathcal{D}, \sigma)$ 
16:   /* Policy Mirror Descent */
17:   let  $E \in \mathbb{R}^{n \times |\mathcal{A}|}$  with rows  $E_i = \text{ONEHOT}_{|\mathcal{A}|}(a_i)$ .
18:   let  $K \in \mathbb{R}^{n \times n}$  such that  $K_{ij} = k(x'_i, x'_j)$ 
19:   let  $H \in \mathbb{R}^{n \times n}$  such that  $H_{ij} = k(x_i, x'_j)$ 
20:   let  $\Phi \in \mathbb{R}^{n \times z_x}$  such that  $\Phi = \{\phi(x_i)\}_{i=1}^n$ 
21:   let  $\Psi \in \mathbb{R}^{n \times z_{xa}}$  such that  $\Psi = \{\phi(x_i) \otimes e_a\}_{i=1}^n$ 
22:   compute  $G = (\Psi \Psi^\top + \text{Id}_n)^{-1}$ 
23:   compute  $\alpha \in \mathbb{R}^n$  such that  $\nu_0 = \Phi^\top \alpha$ 
24:   /* Policy Mirror Descent Updates */
25:   for  $t = 0, 1, \dots, \text{PMD} - 1$  do:
26:      $\pi_{t+1} = \text{SOFTMAX}(-\eta H^\top C_t) \in \mathbb{R}^{n \times |\mathcal{A}|}$  ▷ PMD Step (??)
27:      $M_{\pi_{t+1}} = H \odot (E \pi_{t+1}^\top) \in \mathbb{R}^{n \times n}$  ▷ ??, ??
28:      $S_{\pi_{t+1}} = (\text{Id} - \gamma G M_{\pi_{t+1}})^{-\top} \Phi$ 
29:      $C_{t+1} = C_t + c \odot E$  with  $c = 2\gamma(1 - \gamma)^2 G^\top S_{\pi_{t+1}} S_{\pi_{t+1}}^\top \alpha$ 
30:   end for
31: end procedure

```

1.3.3 Properties

- Se uno stato tabular non viene visto, allora la T sarà nulla. Per minimizzare la loss, l'ottimizzazione porta la massa su quegli stati non visti. Però sembra funzionare con one-hot encoding e non è chiaro come possa funzionare con features imparate.
- Questo da problemi in subsampling, perchè gli stati che non vengono samplati e messi nel kernel porteranno la massa lì. C'è bisogno di un sink state.

1.4 Adding the sink state

1.4.1 The Intuition

In Alg. 1, we optimize—via policy mirror descent—the norm of the discounted state distribution. This corresponds to finding the point on the state simplex that is closest to the origin (Fig. 1).

Minimizing the norm $|\nu_\pi|$ causes the optimization process to shift probability mass toward unknown states, as this reduces the value of the norm. As a consequence, the algorithm exhibits a purely exploratory behavior: it prioritizes the discovery of new states and only subsequently focuses on optimization (exploitation) over the discovered regions.

MPr: Da capire matematicamente

This mechanism relies on retaining all previously observed data in order to correctly compute the kernels in Alg. 1. However, this approach becomes highly inefficient as the number of data points grows.

Moreover, if we restrict the dataset to only the m points sampled from the most recent policy π_t , the algorithm exhibits a “bouncing” behavior between exploring new states and revisiting forgotten ones. To illustrate this phenomenon, we consider a practical experiment. Let the environment consist of ten rooms connected by a single corridor (see Fig. 2). The agent can perform four actions: up, down, left, and right. When implementing the algorithm in its original form and retaining all previously observed data via one-hot embeddings, the agent is able to visit the entire state–action space. As discussed above, this behavior arises because the transition operator redistributes probability mass toward unknown regions in order to minimize ν_π .

MPr: da capire anche analiticamente, che non mi è chiaro

However, maintaining the full dataset is computationally expensive when computing the kernel. From a neural network perspective, each element of the dataset would correspond to a neuron in the first layer, leading to severe scalability issues. This could be addressed by employing subsampling techniques, such as Nystrom. However, the fully exploratory behavior of the original algorithm remains.

To test the effect this limitation, we restrict the dataset to samples generated by the most recent policy π_t during the optimization process. Due to the intrinsically exploratory nature of the induced behavior, the agent does not revisit previously encountered states, but instead continues to explore only novel regions of the state space. This effect is illustrated in Fig. 3: the agent initially explores room 1. Given that states then progressively moves only toward new states. Subsequently, given only the new states, it returns to revisit room 1. This process repeats as a cycle.

Therefore, by introducing an additional sink state, our goal is to encourage the agent to revisit previously explored states, effectively performing a “refresh” of earlier regions. Formally, we define a uniform distribution over the $m \leq |\mathcal{X}|$ visited states augmented with the sink state.

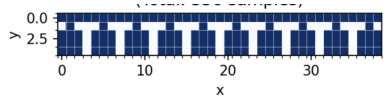


Figure 2: 10 rooms and one corridor environment

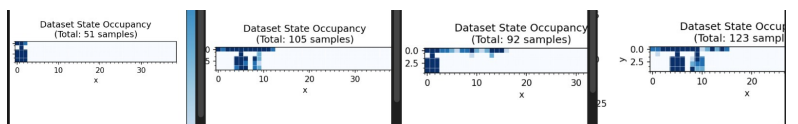


Figure 3: Bouncing effect

1.4.2 Formal Definition

We set $\hat{\Phi} \in \mathbb{R}^{n \times z_x + 1}$, $\hat{\Psi} \in \mathbb{R}^{n \times z_{xa} + 1}$ as the datasets containing the state and the state-action pair embeddings augmented with an additional dimension, which is set to 0. The sink state is defined as $e = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^{z_x + 1}$. In the following analysis, we define $\mathbb{1}_k$ as a column vector containing k -times the 1

value. The augmented version of embedding datasets $\tilde{\Phi}$ and $\tilde{\Psi}$ are defined in the following manner:

MPPr: Da ora $\Psi^*[-1, -1] = 0$, da modificare tutti i calcoli

$$\tilde{\Phi}^* = \begin{bmatrix} \hat{\Phi}^* - e\mathbb{1}_{z_{xa}+1}^* \hat{\Psi}^* & e \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{z_x+2 \times n+1} \quad (86)$$

$$\tilde{\Psi} = \begin{bmatrix} \hat{\Psi} & 0 \\ \mathbb{1}_{z_{xa}+1}^* & 0 \end{bmatrix} \in \mathbb{R}^{n+1 \times z_{xa}+2} \quad (87)$$

The gram-matrix $\hat{B} = (\hat{\Psi}\hat{\Psi}^T + \lambda \text{Id})^{-1}$ is redefined as:

$$\tilde{B} = \begin{bmatrix} \hat{B} & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{n+1 \times n+1}. \quad (88)$$

The initial distribution $\tilde{\nu}_0 = \tilde{\Phi}^* \tilde{\alpha}_0$, where $\tilde{\alpha}_0 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1}$.

1.4.3 Transition to the sink state

The new transition operator T is now $\tilde{\mathsf{T}} = \tilde{\Phi}^* \tilde{B} \tilde{\Psi} \in \mathbb{R}^{z_x+2 \times z_{xa}+2}$. Here, we compute the result of the transition operator applied to the sink state. Before passing through T , it passes through P_π and the outcome is $e_{xa} \in \mathbb{R}^{z_{xa}+1}$.

$$\tilde{\mathsf{T}} \begin{bmatrix} e_{xa} \\ 1 \end{bmatrix} = \tilde{\Phi}^* \tilde{B} \tilde{\Psi} \begin{bmatrix} e_{xa} \\ 1 \end{bmatrix} \quad (89)$$

$$= \tilde{\Phi}^* \tilde{B} \begin{bmatrix} \hat{\Psi} & 0 \\ \mathbb{1}_{z_{xa}+1}^* & 0 \end{bmatrix} \begin{bmatrix} e_{xa} \\ 1 \end{bmatrix} \quad (90)$$

$$= \tilde{\Phi}^* \tilde{B} \begin{bmatrix} \hat{\Psi} e_{xa} \\ \mathbb{1}_{z_{xa}+1}^* e \end{bmatrix} \quad (91)$$

$$= \tilde{\Phi}^* \begin{bmatrix} \hat{B} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (92)$$

$$= \tilde{\Phi}^* \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (93)$$

$$= \begin{bmatrix} \hat{\Phi}^* - e\mathbb{1}_{z_{xa}+1}^* \hat{\Psi}^* & e \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (94)$$

$$= \begin{bmatrix} e \\ 1 \end{bmatrix} \quad (95)$$

Here, we showed that the transition computed on a sink state leads again to the sink state. In 91, we note that $\hat{\Psi}e_a = 0$ by construction, since the two terms are orthogonal and $\mathbb{1}_{z_{xa}+1}^* e = 1$.

1.4.4 Transition to a state

Now, we suppose to transition to a state, then:

MPPr: sto mettendo $\hat{\Psi}$

$$\tilde{T} \begin{bmatrix} \hat{\psi}(x) \\ 1 \end{bmatrix} = \tilde{\Phi}^* \tilde{B} \tilde{\Psi} \begin{bmatrix} \hat{\psi}(x) \\ 1 \end{bmatrix} \quad (96)$$

$$= \tilde{\Phi}^* \tilde{B} \begin{bmatrix} \hat{\Psi} & 0 \\ \mathbb{1}_{z_{xa}+1}^* & 0 \end{bmatrix} \begin{bmatrix} \hat{\psi}(x) \\ 1 \end{bmatrix} \quad (97)$$

$$= \tilde{\Phi}^* \begin{bmatrix} \hat{B} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\Psi} \hat{\psi}(x, a) \\ \mathbb{1}_{z_{xa}+1}^* \hat{\psi}(x, a) \end{bmatrix} \quad (98)$$

$$= \begin{bmatrix} \hat{\Phi}^* - e \mathbb{1}_{z_{xa}+1}^* \hat{\Psi}^* & e \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{B} \hat{\Psi} \hat{\psi}(x, a) \\ \mathbb{1}_{z_{xa}+1}^* \hat{\psi}(x, a) \end{bmatrix} \quad (99)$$

$$= \begin{bmatrix} \Phi^* B \Psi \psi(x, a) - e \mathbb{1}_{z_{xa}+1}^* \Psi^* B \Psi \psi(x, a) + e \mathbb{1}_{z_{xa}+1}^* \psi(x, a) \\ \mathbb{1}_{z_{xa}+1}^* \psi(x, a) \end{bmatrix} = (1), (2) \quad (100)$$

$$(1) \approx \begin{bmatrix} \Phi^* B \Psi \psi(x, a) + e \mathbb{1}_{z_{xa}+1}^* \psi(x, a) \\ 1 \end{bmatrix}, \text{ if } \Psi \psi(x, a) \approx 0 \quad (101)$$

$$(2) \approx \begin{bmatrix} \Phi^* B \Psi \psi(x, a) \\ 1 \end{bmatrix} \quad (102)$$

In 100, $B = (\Psi \Psi^T + \lambda \text{Id})^{-1} \approx (\Psi \Psi^T)^{-1}$ for a small λ . Then, $e \mathbb{1}_{z_{xa}+1}^* \Psi^* B \Psi \psi(x, a) \approx e \mathbb{1}_{z_{xa}+1}^* \psi(x, a)$ and it leads to 102. In the end, let $\mathcal{H}$ be an RKHS over $\mathcal{X} \times \mathcal{A}$ with feature map $\psi : (x, a) \mapsto \psi(x, a) \in \mathcal{H}$. We assume that the constant-one function belongs to $\mathcal{H}$, i.e., there exists $\mathbf{1}_{\mathcal{H}} \in \mathcal{H}$ such that

$$\langle \mathbf{1}_{\mathcal{H}}, \psi(x, a) \rangle_{\mathcal{H}} = 1, \quad \forall (x, a) \in \mathcal{X} \times \mathcal{A}. \quad (103)$$

1.4.5 The M matrix

In 61, we defined $M = \Psi P \Phi^T$. Now, we redefine $\tilde{M} = \tilde{\Psi} \tilde{P} \tilde{\Phi}^T$, where $\tilde{P} = \begin{bmatrix} P & 0 \\ 0 & 0 \end{bmatrix}$

MP: diversa dalla M = TP iniziale che è da cambiare

$$\tilde{M} = \tilde{\Psi} \tilde{P} \tilde{\Phi}^T \quad (104)$$

$$= \tilde{\Psi} \begin{bmatrix} P & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \Phi^* - e \mathbb{1}_{z_{xa}+1}^* \Psi^* & e \\ 0 & 1 \end{bmatrix} \quad (105)$$

$$= \tilde{\Psi} \begin{bmatrix} P \Phi^* - P e \mathbb{1}_{z_{xa}+1}^* \Psi^* & P e \\ 0 & 0 \end{bmatrix} \quad (106)$$

$$= \begin{bmatrix} \Psi & 0 \\ \mathbb{1}_{z_{xa}+1}^* & 0 \end{bmatrix} \begin{bmatrix} P \Phi^* - P e \mathbb{1}_{z_{xa}+1}^* \Psi^* & P e \\ 0 & 0 \end{bmatrix} \quad (107)$$

$$= \begin{bmatrix} \Psi P \Phi^* - \Psi P e \mathbb{1}_{z_{xa}+1}^* \Psi^* & \Psi P e \\ \mathbb{1}_{z_{xa}+1}^* (P \Phi^* - P e \mathbb{1}_{z_{xa}+1}^* \Psi^*) & \mathbb{1}_{z_{xa}+1}^* P e \end{bmatrix} \quad (108)$$

$$(109)$$

We analyze the M matrix point-wise, starting from the upper-left block.

$$\tilde{M}_{ij} = \psi(x_i, a_i) (P \phi^*(x'_j) - P e \mathbb{1}_{z_{xa}+1}^* \psi^*(x_j, a_j)) \quad (110)$$

$$\langle \psi(x_i, a_i)^*, P \phi(x'_j) \rangle = \langle \phi(x_i) \otimes e_{a_i}, \sum_{a \in \mathcal{A}} \pi(a | x'_j) \phi(x'_j) \otimes e_a \rangle \quad (111)$$

$$= \sum_{a \in \mathcal{A}} \pi(a | x'_j) \langle \phi(x_i), \phi(x'_j) \rangle \otimes e_{a=a_i} \quad (112)$$

$$= \pi(a | x'_j) K(x_i, x'_j) \quad (113)$$

$$\psi(x_i, a_i) P e \mathbb{1}_{z_{xa}+1}^* \psi(x_j, a_j) = \pi(a_i | e) K(x_i, e) \mathbb{1}_{z_{xa}+1}^* \psi(x_j, a_j) = 0 \quad (114)$$

$$\tilde{M}_{ij} = \pi(a | x'_j) K(x_i, x'_j) = M_{ij} \quad (115)$$

The upper-right block yields to $\pi(a_i|e)K(x_i, e) = 0$ as in 114, which is 0 due to $K(x_i, e)$. Moreover, the bottom-left derivation is as follows:

$$\tilde{M}_i = \mathbb{1}_{z_{x_a}+1}^* (\mathbf{P}\phi^*(x'_i) - \mathbf{P}e\mathbb{1}_{z_{x_a}+1}^* \psi^*(x_i, a_i)) \quad (116)$$

$$\langle \mathbb{1}_{z_{x_a}+1}, \mathbf{P}\phi^*(x'_i) \rangle = \langle \mathbb{1}_{z_{x_a}+1}, \pi(\cdot|x'_i) \otimes \phi(x'_i) \rangle \quad (117)$$

$$= \sum_{a \in \mathcal{A}} \pi(a|x'_i) \langle \mathbb{1}_{z_{x_a}+1}, \phi(x'_i) \otimes \delta_a \rangle \quad (118)$$

$$= \sum_{a \in \mathcal{A}} \pi(a|x'_i) \langle \mathbb{1}_{z_{x_a}+1}, \psi(x'_i, a) \rangle = 1 \quad (119)$$

$$\mathbb{1}_{z_{x_a}+1}^* \mathbf{P}e\mathbb{1}_{z_{x_a}+1}^* \psi^*(x_i, a_i) = \sum_{a \in \mathcal{A}} \pi(a|e) \langle \mathbb{1}_{z_{x_a}+1}, e \otimes e_a \rangle = 1 \quad (120)$$

$$\tilde{M}_i = 0 \quad (121)$$

$$\tilde{M} = \begin{bmatrix} M & 0 \\ 0 & 1 \end{bmatrix} \quad (122)$$

1.4.6 The gradient

Here, we plug the tlide elements computed in the sections above into the gradient of Eq. 61:

$$\nabla_{\mathbf{P}_\pi} \text{MMD} = \beta \tilde{\Psi}^\top \tilde{B}^\top (\text{Id} - \gamma \tilde{B} \tilde{M})^{-\top} \tilde{\Phi} \tilde{\Phi}^\top (\text{Id} - \gamma \tilde{B} \tilde{M})^{-1} \tilde{\alpha} \quad (123)$$

$$= \beta \tilde{\Psi}^\top \begin{bmatrix} B^\top & 0 \\ 0 & 1 \end{bmatrix} (\text{Id} - \gamma \begin{bmatrix} B & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} M & 0 \\ 0 & 1 \end{bmatrix})^{-\top} \tilde{\Phi} \tilde{\Phi}^\top (\text{Id} - \gamma \begin{bmatrix} B & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} M & 0 \\ 0 & 1 \end{bmatrix})^{-1} \tilde{\alpha} \quad (124)$$

$$= \beta \tilde{\Psi}^\top \begin{bmatrix} B^\top & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} (\text{Id}_n - \gamma BM)^{-\top} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \tilde{\Phi} \tilde{\Phi}^\top \begin{bmatrix} (\text{Id}_n - \gamma BM)^{-1} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \tilde{\alpha} \quad (125)$$

$$= \beta \tilde{\Psi}^\top \begin{bmatrix} B^\top (\text{Id}_n - \gamma BM)^{-\top} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \begin{bmatrix} \Phi - \Psi \mathbb{1}_{z_{xa}+1} e^* & 0 \\ e^* & 1 \end{bmatrix} \begin{bmatrix} \Phi^* - e \mathbb{1}_{z_{xa}+1}^* \Psi^* & e \\ 0 & 1 \end{bmatrix} \quad (126)$$

$$\begin{bmatrix} (\text{Id}_n - \gamma BM)^{-1} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \tilde{\alpha} \quad (127)$$

$$= \beta \tilde{\Psi}^\top \begin{bmatrix} B^\top (\text{Id}_n - \gamma BM)^{-\top} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \quad (128)$$

$$\begin{bmatrix} \Phi \Phi^* - \cancel{\Phi e \mathbb{1}_{z_{xa}+1}^* \Psi^*} - \cancel{\Psi \mathbb{1}_{z_{xa}+1} e^* \Phi^*} + \Psi \mathbb{1}_{z_{xa}+1} e^* e \mathbb{1}_{z_{xa}+1}^* \Psi^* & \cancel{\Phi e} - \Psi \mathbb{1}_{z_{xa}+1} e^* e \\ \cancel{e^* \Phi^*} - \cancel{e^* e \mathbb{1}_{z_{xa}+1}^* \Psi^*} & e^* e + 1 \end{bmatrix} \quad (129)$$

$$\begin{bmatrix} (\text{Id}_n - \gamma BM)^{-1} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \tilde{\alpha} \quad (130)$$

$$= \beta \tilde{\Psi}^\top \begin{bmatrix} B^\top (\text{Id}_n - \gamma BM)^{-\top} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \begin{bmatrix} \Phi \Phi^* + \mathbb{1}_n \mathbb{1}_n^* & -\mathbb{1}_n \\ -\mathbb{1}_n^* & 2 \end{bmatrix} \quad (131)$$

$$\begin{bmatrix} (\text{Id}_n - \gamma BM)^{-1} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \tilde{\alpha} \quad (132)$$

$$= \beta \tilde{\Psi}^\top \begin{bmatrix} B^\top (\text{Id}_n - \gamma BM)^{-\top} (\Phi \Phi^* + \mathbb{1}_n \mathbb{1}_n^*) & -B^\top (\text{Id}_n - \gamma BM)^{-\top} \mathbb{1}_n \\ -(1 - \gamma)^{-1} \mathbb{1}_n^* & 2(1 - \gamma)^{-1} \end{bmatrix} \quad (133)$$

$$\begin{bmatrix} (\text{Id}_n - \gamma BM)^{-1} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \tilde{\alpha} \quad (134)$$

$$= \beta \begin{bmatrix} \Psi^* & \mathbb{1}_{z_{xa}+1} \\ 0 & 0 \end{bmatrix} \quad (135)$$

$$\begin{bmatrix} B^\top (\text{Id}_n - \gamma BM)^{-\top} (\Phi \Phi^* + \mathbb{1}_n \mathbb{1}_n^*) (\text{Id}_n - \gamma BM)^{-1} & -(1 - \gamma)^{-1} B^\top (\text{Id}_n - \gamma BM)^{-\top} \mathbb{1}_n \\ -(1 - \gamma)^{-1} \mathbb{1}_n^* (\text{Id}_n - \gamma BM)^{-1} & 2(1 - \gamma)^{-2} \end{bmatrix} \tilde{\alpha} \quad (136)$$

$$= \beta \begin{bmatrix} \Psi^* B^\top (\text{Id}_n - \gamma BM)^{-\top} (\Phi \Phi^* + \mathbb{1}_n \mathbb{1}_n^*) (\text{Id}_n - \gamma BM)^{-1} - (1 - \gamma)^{-1} \mathbb{1}_{z_{xa}+1} \mathbb{1}_n^* (\text{Id}_n - \gamma BM)^{-1} & \\ -(1 - \gamma)^{-1} \Psi^* B^\top (\text{Id}_n - \gamma BM)^{-\top} \mathbb{1}_n + 2(1 - \gamma)^{-2} \mathbb{1}_{z_{xa}+1} & \\ 0 & 0 \end{bmatrix} \tilde{\alpha} \quad (137)$$

$$\begin{bmatrix} \psi(x, a) \\ 1 \end{bmatrix}^\top \begin{bmatrix} \Psi^* & \mathbb{1}_{z_{xa}+1} \\ 0 & 0 \end{bmatrix} C \quad (138)$$

$$\begin{bmatrix} \psi^*(x, a) \Psi^* & 1 \end{bmatrix} C \quad (139)$$

$$\nu_{\mathbf{P}_\pi} = (1 - \gamma) \Phi^* (\text{Id} - \gamma BM)^{-1} \alpha \quad (140)$$

$$= (1 - \gamma) \begin{bmatrix} \hat{\Phi}^* - e \mathbb{1}_{z_{xa}+1}^* \hat{\Psi}^* & e \\ 0 & 1 \end{bmatrix} (\text{Id} - \gamma \begin{bmatrix} BM & 0 \\ 0 & 1 \end{bmatrix})^{-1} \tilde{\alpha} \quad (141)$$

$$\nu_\pi(e) = (1 - \gamma) \begin{bmatrix} e^* & 0 \end{bmatrix} \begin{bmatrix} \hat{\Phi}^* - e \mathbb{1}_{z_{xa}+1}^* \hat{\Psi}^* & e \\ 0 & 1 \end{bmatrix} \begin{bmatrix} (\text{Id} - \gamma BM)^{-1} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \tilde{\alpha} \quad (142)$$

$$= (1 - \gamma) \begin{bmatrix} -\mathbb{1}_{z_n}^* & 1 \end{bmatrix} \begin{bmatrix} (\text{Id} - \gamma BM)^{-1} & 0 \\ 0 & (1 - \gamma)^{-1} \end{bmatrix} \tilde{\alpha} \quad (143)$$

$$= (1 - \gamma) \begin{bmatrix} -\mathbb{1}_{z_n}^* (\text{Id} - \gamma BM)^{-1} & (1 - \gamma)^{-1} \end{bmatrix} \tilde{\alpha} \quad (144)$$

$$(145)$$

1.4.7 The Modified Update Rule

As in Sec. 1.3.1, we want to derive the update rule considering the new definitions of $\tilde{\Psi}$ and $\tilde{\Phi}$. Let $\tilde{c} \in \mathbb{R}^{n+1}$ be an arbitrary coefficient vector. We expand the quantity $\tilde{\Psi}^\top \tilde{c}$ as:

$$\tilde{\Psi}^\top \tilde{c} = \begin{bmatrix} \Psi^* & \mathbb{1}_{z_{xa}+1} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{c}_{:n} \\ \tilde{c}_{n+1} \end{bmatrix} \quad (146)$$

$$= \begin{bmatrix} \Psi^* \tilde{c}_{:n} + \mathbb{1}_{z_{xa}+1} \tilde{c}_{n+1} \\ 0 \end{bmatrix} \in \mathbb{R}^{z_{xa}+2} \quad (147)$$

Where $\tilde{c}_n$ are all the elements up to n and $\tilde{c}_{n+1}$ is the scalar in the last cell. At this point, we extend the analysis by discarding the null term. Follows:

$$\Psi^* \tilde{c}_{:n} + \mathbb{1}_{z_{xa}+1} \tilde{c}_{n+1} = \sum_{i=0}^n \tilde{c}_i \phi(x_i) \otimes \delta_{a_i} + \begin{bmatrix} \tilde{c}_{n+1} \\ \vdots \\ \tilde{c}_{n+1} \end{bmatrix} \quad (148)$$

As in Sec. 1.3.1, we reshape this block $\Psi^* \tilde{c}_{:n} + \mathbb{1}_{z_{xa}+1} \tilde{c}_{n+1} \in \mathbb{R}^{z_{xa}+1}$ into a matrix in $\mathbb{R}^{z_x+1 \times |\mathcal{A}|}$, where each column corresponds to one action:

$$\text{mat}(\Psi^* \tilde{c}_{:n} + \mathbb{1}_{z_{xa}+1} \tilde{c}_{n+1}) = \underbrace{\left[\sum_{i: a_i=1} \tilde{c}_i \phi(x_i) + \mathbb{1}_{z_x+1} \tilde{c}_{n+1} \right]}_{\in \mathbb{R}^{z_x+1}} \underbrace{\left[\sum_{i: a_i=2} \tilde{c}_i \phi(x_i) + \mathbb{1}_{z_x+1} \tilde{c}_{n+1} \right]}_{\in \mathbb{R}^{z_x+1}} \cdots \underbrace{\left[\sum_{i: a_i=A} \tilde{c}_i \phi(x_i) + \mathbb{1}_{z_x+1} \tilde{c}_{n+1} \right]}_{\in \mathbb{R}^{z_x+1}}. \quad (149)$$

Stacking all columns together yields the fully compact expression

$$\boxed{\text{mat}(\tilde{\Psi}^\top \tilde{c}) = \Phi^\top (\tilde{c}_{:n} \odot E) + \mathbb{1}_{z_x+1} \tilde{c}_{n+1} \mathbb{1}_{|\mathcal{A}|}^* \in \mathbb{R}^{z_x+1 \times |\mathcal{A}|}.} \quad (150)$$

1.5 Gaussian Kernel

Let w be the weights on observed transitions, and assume the normalized state-action features used in the sink construction sum to one. The paper's actual augmented features are $\bar{\phi}(x) = (\phi(x), 0)$ and $\bar{\phi}(e) = (0, \varepsilon)$. Its shifted estimator uses residuals $r_i = \bar{\phi}(x'_i) - \bar{\phi}(e) = (\phi(x'_i), -\varepsilon)$, together with a baseline sink feature $\bar{\phi}(e)$, so the occupancy embedding is $\sum_i w_i r_i + \bar{\phi}(e) = \sum_i w_i \phi(x'_i) + (1 - \mathbf{1}^\top w) \bar{\phi}(e)$. With an inner product, the residual Gram blocks are $K_{rr} = K_{XY} + \varepsilon^2 \mathbf{1} \mathbf{1}^\top$, $K_{re} = -\varepsilon^2 \mathbf{1}$, and $K_{ee} = \varepsilon^2$; together they give $J = w^\top K_{XY} w + \varepsilon^2 (1 - \mathbf{1}^\top w)^2$, as required by the orthogonal sink construction ROVER, Sec. 3.1. An RBF kernel applied directly to the residual coordinates does not preserve these inner products: for $r_i = (x_i, -\varepsilon)$ and $e = (0, +\varepsilon)$, it gives $K_{rr} = K_{XY}$, $K_{re,i} = \exp[-(\|x_i\|^2 + 4\varepsilon^2)/(2\sigma^2)]$, and $K_{ee} = 1$. The resulting objective contains a location-dependent state-sink term and makes the sink's self-kernel independent of ε ; this can make distant states cheaper and bias coverage toward outer regions. Merely changing the state coordinate to $(x, y, 0)$ while applying the same RBF is also insufficient: the state-sink kernel remains positive and the sink self-kernel remains one. A Gaussian coverage kernel therefore needs an explicitly orthogonal sink block, or an equivalent residual Gram construction.

1.6 Nystrom

1.6.1 Projected Nyström gradient

We compute the projected gradient

$$\tilde{g}_{\text{proj}} = \beta \tilde{B}_r^\top \tilde{S}_{r\varepsilon}^{-\top} \tilde{K}_{\phi,r} \tilde{S}_r^{-1} \tilde{\alpha}_r.$$

Here, **we do not expand the internal structure of $\tilde{K}_{\phi,r}$, since the kernel may be a gaussian kernel.** Instead, we only use its block decomposition

$$\tilde{K}_{\phi,r} = \begin{bmatrix} K_{rr} & k_{re} \\ k_{er}^\top & k_{ee} \end{bmatrix},$$

where

$$K_{rr} \in \mathbb{R}^{r \times r}, \quad k_{re} \in \mathbb{R}^r, \quad k_{er} \in \mathbb{R}^r, \quad k_{ee} \in \mathbb{R}.$$

If $\tilde{K}_{\phi,r}$ is symmetric, then $k_{er} = k_{re}$, but we do not need to impose this in the computation.

Let

$$s := 1 - \gamma,$$

and write

$$\tilde{S}_r = \begin{bmatrix} S_r & 0 \\ 0 & s \end{bmatrix}, \quad \tilde{S}_{r\varepsilon} = \begin{bmatrix} S_r + \varepsilon I_r & 0 \\ 0 & s + \varepsilon \end{bmatrix}, \quad \tilde{B}_r = \begin{bmatrix} B_r & 0 \\ 0 & 1 \end{bmatrix},$$

with

$$S_r \in \mathbb{R}^{r \times r}, \quad B_r \in \mathbb{R}^{r \times r}.$$

Moreover, let

$$\tilde{\alpha}_r = \begin{bmatrix} \alpha_r \\ \alpha_e \end{bmatrix}, \quad \alpha_r \in \mathbb{R}^r, \quad \alpha_e \in \mathbb{R}.$$

We now multiply the terms from right to left. First,

$$\tilde{S}_r^{-1} \tilde{\alpha}_r = \begin{bmatrix} S_r^{-1} & 0 \\ 0 & s^{-1} \end{bmatrix} \begin{bmatrix} \alpha_r \\ \alpha_e \end{bmatrix} = \begin{bmatrix} S_r^{-1} \alpha_r \\ s^{-1} \alpha_e \end{bmatrix}.$$

Define

$$z_r := S_r^{-1} \alpha_r, \quad z_e := s^{-1} \alpha_e.$$

Then

$$\tilde{S}_r^{-1} \tilde{\alpha}_r = \begin{bmatrix} z_r \\ z_e \end{bmatrix}.$$

Next, multiplying by $\tilde{K}_{\phi,r}$, we obtain

$$\tilde{K}_{\phi,r} \tilde{S}_r^{-1} \tilde{\alpha}_r = \begin{bmatrix} K_{rr} & k_{re} \\ k_{er}^\top & k_{ee} \end{bmatrix} \begin{bmatrix} z_r \\ z_e \end{bmatrix},$$

hence

$$\tilde{K}_{\phi,r} \tilde{S}_r^{-1} \tilde{\alpha}_r = \begin{bmatrix} K_{rr} z_r + k_{re} z_e \\ k_{er}^\top z_r + k_{ee} z_e \end{bmatrix}.$$

Substituting back $z_r = S_r^{-1} \alpha_r$ and $z_e = s^{-1} \alpha_e$, this becomes

$$\tilde{K}_{\phi,r} \tilde{S}_r^{-1} \tilde{\alpha}_r = \begin{bmatrix} K_{rr} S_r^{-1} \alpha_r + k_{re} s^{-1} \alpha_e \\ k_{er}^\top S_r^{-1} \alpha_r + k_{ee} s^{-1} \alpha_e \end{bmatrix}.$$

Now we multiply by $\tilde{S}_{r\varepsilon}^{-\top}$:

$$\tilde{S}_{r\varepsilon}^{-\top} \tilde{K}_{\phi,r} \tilde{S}_r^{-1} \tilde{\alpha}_r = \begin{bmatrix} (S_r + \varepsilon I_r)^{-\top} & 0 \\ 0 & (s + \varepsilon)^{-1} \end{bmatrix} \begin{bmatrix} K_{rr} S_r^{-1} \alpha_r + k_{re} s^{-1} \alpha_e \\ k_{er}^\top S_r^{-1} \alpha_r + k_{ee} s^{-1} \alpha_e \end{bmatrix}.$$

Therefore,

$$\tilde{S}_{r\varepsilon}^{-\top} \tilde{K}_{\phi,r} \tilde{S}_r^{-1} \tilde{\alpha}_r = \begin{bmatrix} (S_r + \varepsilon I_r)^{-\top} [K_{rr} S_r^{-1} \alpha_r + k_{re} s^{-1} \alpha_e] \\ (s + \varepsilon)^{-1} [k_{er}^\top S_r^{-1} \alpha_r + k_{ee} s^{-1} \alpha_e] \end{bmatrix}.$$

Finally, multiplying by $\tilde{B}_r^\top$, we obtain

$$\tilde{g}_{\text{proj}} = \beta \begin{bmatrix} B_r^\top & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} (S_r + \varepsilon I_r)^{-\top} [K_{rr} S_r^{-1} \alpha_r + k_{re} s^{-1} \alpha_e] \\ (s + \varepsilon)^{-1} [k_{er}^\top S_r^{-1} \alpha_r + k_{ee} s^{-1} \alpha_e] \end{bmatrix}.$$

Thus,

$$\tilde{g}_{\text{proj}} = \beta \begin{bmatrix} B_r^\top (S_r + \varepsilon I_r)^{-\top} [K_{rr} S_r^{-1} \alpha_r + k_{re} s^{-1} \alpha_e] \\ (s + \varepsilon)^{-1} [k_{er}^\top S_r^{-1} \alpha_r + k_{ee} s^{-1} \alpha_e] \end{bmatrix}.$$

The lifted gradient in the augmented Nyström space is

$$\tilde{g} = \tilde{U}_r \tilde{g}_{\text{proj}} = \begin{bmatrix} U_r & 0 \\ 0 & 1 \end{bmatrix} \tilde{g}_{\text{proj}}.$$

Therefore,

$$\tilde{g} = \beta \begin{bmatrix} U_r B_r^\top (S_r + \varepsilon I_r)^{-\top} [K_{rr} S_r^{-1} \alpha_r + k_{re} s^{-1} \alpha_e] \\ (s + \varepsilon)^{-1} [k_{er}^\top S_r^{-1} \alpha_r + k_{ee} s^{-1} \alpha_e] \end{bmatrix}.$$

1.7 Inductive Bias - scritto al volo con gpt da scrivere meglio

We distinguish between the geometry required to model the dynamics and the geometry in which state-space coverage is measured. Let $\mathcal{X}$ denote the Markov state space and let

$$g : \mathcal{X} \rightarrow \mathcal{Z}$$

be a measurable map selecting the subspace of interest. We introduce a *dynamics kernel*

$$k_d(x, \tilde{x}) = \langle \phi_d(x), \phi_d(\tilde{x}) \rangle_{\mathcal{H}_d},$$

which retains all information required to model the transition dynamics, and a *coverage kernel*

$$k_c(x, \tilde{x}) = \kappa(g(x), g(\tilde{x})) = \langle \phi_c(x), \phi_c(\tilde{x}) \rangle_{\mathcal{H}_c}, \quad \phi_c(x) := \zeta(g(x)),$$

which measures similarity only in the subspace of interest. For example, in PointMaze,

$$x = (p_x, p_y, v_x, v_y), \quad g(x) = (p_x, p_y).$$

Two possible Gaussian kernels are

$$k_d(x, \tilde{x}) = \exp \left(-\frac{1}{2} (x - \tilde{x})^\top \Lambda_d^{-1} (x - \tilde{x}) \right)$$

and

$$k_c(x, \tilde{x}) = \exp \left(-\frac{(p_x - \tilde{p}_x)^2 + (p_y - \tilde{p}_y)^2}{2\sigma_c^2} \right).$$

Hence, k_d distinguishes states with different velocities whenever this is relevant for predicting their future evolution, whereas k_c identifies states having the same position, independently of their velocities.

Given transitions $\{(x_i, a_i, x'_i)\}_{i=1}^n$, define the dynamics state-action feature

$$\psi_d(x, a) = \phi_d(x) \otimes e_a,$$

and the feature matrices

$$\Phi_d = \begin{bmatrix} \phi_d(x'_1)^\top \\ \vdots \\ \phi_d(x'_n)^\top \end{bmatrix}, \quad \Psi_d = \begin{bmatrix} \psi_d(x_1, a_1)^\top \\ \vdots \\ \psi_d(x_n, a_n)^\top \end{bmatrix}.$$

The conditional mean embedding of the full-state transition kernel is estimated as

$$\widehat{T} = \Phi_d^\top G \Psi_d, \quad G = (\Psi_d \Psi_d^\top + \lambda I)^{-1}.$$

Let

$$M_\pi = \Psi_d P_\pi \Phi_d^\top, \quad \mu_{d,0} = \Phi_d^\top \alpha_0.$$

The discounted full-state occupancy embedding is then

$$\widehat{\mu}_{d,\pi} = (1 - \gamma) \left(I - \gamma \widehat{T} P_\pi \right)^{-1} \mu_{d,0} \quad (151)$$

$$= (1 - \gamma) \Phi_d^\top (I - \gamma G M_\pi)^{-1} \alpha_0. \quad (152)$$

Defining

$$\beta_\pi := (I - \gamma G M_\pi)^{-1} \alpha_0,$$

the same occupancy coefficients can be used to embed the occupancy of the subspace of interest. In particular, if

$$\Phi_c = \begin{bmatrix} \phi_c(x'_1)^\top \\ \vdots \\ \phi_c(x'_n)^\top \end{bmatrix}, \quad K_c = \Phi_c \Phi_c^\top,$$

then

$$\widehat{\mu}_{c,\pi} = (1 - \gamma) \Phi_c^\top \beta_\pi.$$

This is the kernel mean embedding of the projected occupancy

$$q^\pi := g_\# d^\pi, \quad q^\pi(B) = d^\pi(g^{-1}(B)), \quad B \subseteq \mathcal{Z}.$$

The corresponding coverage objective is

$$\widehat{\mathcal{J}}_c(\pi) = \|\widehat{\mu}_{c,\pi}\|_{\mathcal{H}_c}^2 = (1 - \gamma)^2 \beta_\pi^\top K_c \beta_\pi.$$

Differentiating this objective through the full-state resolvent gives the dual gradient coefficient

$$\boxed{c_\pi^{(c)} = 2\gamma(1 - \gamma)^2 G (I - \gamma M_\pi^\top G)^{-1} K_c (I - \gamma G M_\pi)^{-1} \alpha_0.}$$

Thus, the dynamics kernel determines how occupancy propagates through the environment, while the coverage kernel determines which components of that occupancy are rewarded as coverage.

Finally, the policy gradient is queried using the dynamics state-action feature:

$$\left[\nabla_\pi \widehat{\mathcal{J}}_c(\pi) \right] (x, a) = \left\langle \psi_d(x, a), \Psi_d^\top c_\pi^{(c)} \right\rangle.$$

Equivalently, defining the state-action dynamics kernel

$$k_d^A((x, a), (\tilde{x}, \tilde{a})) = k_d(x, \tilde{x}) \mathbf{1}\{a = \tilde{a}\},$$

the query can be written entirely in kernel form as

$$\left[\nabla_\pi \widehat{\mathcal{J}}_c(\pi) \right] (x, a) = \sum_{i=1}^n [c_\pi^{(c)}]_i k_d^A((x, a), (x_i, a_i)).$$

Consequently, a policy mirror-descent step takes the form

$$\pi_{t+1}(a \mid x) \propto \pi_t(a \mid x) \exp \left(-\eta \sum_{i=1}^n [c_{\pi_t}^{(c)}]_i k_d^A((x, a), (x_i, a_i)) \right).$$

2 Gaussian Policy for Continuous State Space (DRAFT)

Let

$$\pi(a|x) = \mathcal{N}(a; \mu_{\theta_\mu}(x), \Sigma_{\theta_\sigma}(x)) \quad (153)$$

The embedding $\psi(s, a) = \phi(s) \otimes h(a)$, where $h(a)$ is an embedding on the actions and $k_h(a_1, a_2)$ is a gaussian kernel with parameter σ_h . Then, we want to compute the gradient of $f(\theta_\mu, \theta_\sigma) = \|(1 - \gamma)(\text{Id} - \gamma \mathbb{T} \mathbb{P}_\pi)^{-1} \nu\|^2$ wrt the mean and std deviation parameters. For now, we tackle a single action dimension, i.e. $a \in \mathbb{R}^{d_a}$.

$$\partial f = 2\langle m, \partial m \rangle \quad (154)$$

$$= 2\gamma(1 - \gamma) \langle R_\pi \nu, (1 - \gamma) R_\pi \mathbb{T} \partial \mathbb{P}_\pi R_\pi \nu \rangle \quad (155)$$

$$= 2\gamma(1 - \gamma) \langle \mathbb{T}^* R_\pi^* R_\pi \nu, (1 - \gamma) \partial \mathbb{P}_\pi R_\pi \nu \rangle \quad (156)$$

where,

$$(1 - \gamma) \mathbb{P}_\pi R_\pi \nu = \int_{\mathcal{X}} \int_{\mathcal{A}} \psi(x, a) \mathcal{N}(da | \mu_{\theta_\mu}(x), \Sigma_{\theta_\sigma}(x)) d_\pi(dx) \quad (157)$$

$$\partial(1 - \gamma) \mathbb{P}_\pi R_\pi \nu = \int_{\mathcal{X}} \int_{\mathcal{A}} \psi(x, a) \nabla_\theta (\mathcal{N}(da | \mu_{\theta_\mu}(x), \Sigma_{\theta_\sigma}(x)))^\top d_\pi(dx) \partial_\theta \quad (158)$$

And (158) in (156):

$$\partial f = 2\gamma(1 - \gamma) \langle \mathbb{T}^* R_\pi^* R_\pi \nu, \int_{\mathcal{X}} \int_{\mathcal{A}} \psi(x, a) \nabla_\theta (\mathcal{N}(da | \mu_{\theta_\mu}(x), \Sigma_{\theta_\sigma}(x)))^\top d_\pi(dx) \partial_\theta \rangle \quad (159)$$

$$= 2\gamma(1 - \gamma) \int_{\mathcal{X}} \int_{\mathcal{A}} \langle \mathbb{T}^* R_\pi^* R_\pi \nu, \psi(x, a) \rangle \nabla_\theta (\mathcal{N}(da | \mu_{\theta_\mu}(x), \Sigma_{\theta_\sigma}(x)))^\top d_\pi(dx) \partial_\theta \quad (160)$$

We focus on further developments on $\langle \psi(x, a), \mathbb{T}^* R_\pi^* R_\pi \nu \rangle = \langle \mathbb{T}^* R_\pi^* R_\pi \nu, \psi(x, a) \rangle \in \mathbb{R}$, recalling that $\mathbb{T}^* = \Psi^\top G^\top \Phi$. However,

$$\langle \psi(x, a), \mathbb{T}^* R_\pi^* R_\pi \nu \rangle = \langle \psi(x, a), \Psi^\top G^\top \Phi R_\pi^* R_\pi \nu \rangle \quad (161)$$

$$= \langle \psi(x, a), \Psi^\top c_\pi \rangle \quad (162)$$

$$= \langle \phi(x) \otimes h(a), \begin{bmatrix} \phi(x_1) \otimes h(a_1)^\top \\ \phi(x_2) \otimes h(a_2)^\top \\ \vdots \\ \phi(x_n) \otimes h(a_n)^\top \end{bmatrix} c_\pi \rangle \quad (163)$$

$$= (K_\phi(x, x_i) \odot k_h(a, a_i))^\top c_\pi \quad (164)$$

$$= (c_\pi \odot K_\phi(x, x_i))^\top k_h(a, a_i) \quad (165)$$

Now we redefine $(c_\pi \odot K_\phi(x, x_i)) = b_\pi$ and $\xi = 2\gamma(1 - \gamma)$

$$\partial f = \xi \int_{\mathcal{X}} b_\pi(x)^\top \int_{\mathcal{A}} k_h(a, a_i) \nabla_\theta (\mathcal{N}(da | \mu_{\theta_\mu}(x), \Sigma_{\theta_\sigma}(x)))^\top d_\pi(dx) \partial_\theta \quad (166)$$

$$= \xi \int_{\mathcal{X}} b_\pi(x)^\top \int_{\mathcal{A}} k_h(a, a_i) \langle \nabla_\theta (\mathcal{N}(da | \mu_{\theta_\mu}(x), \Sigma_{\theta_\sigma}(x))) d_\pi(dx), \partial_\theta \rangle \quad (167)$$

Then, the gradient $\partial f = \langle \nabla_\theta f, \partial \rangle$

$$\nabla_\theta f = \xi \int_{\mathcal{X}} b_\pi(x)^\top \int_{\mathcal{A}} k_h(a, a_i) \nabla_\theta (\mathcal{N}(da | \mu_{\theta_\mu}(x), \Sigma_{\theta_\sigma}(x))) d_\pi(dx) \quad (168)$$

$$= \xi \mathbb{E}_{x \sim d_\pi} \left[b_\pi(x)^\top \nabla_\theta \int_{\mathcal{A}} k_h(a, a_i) \mathcal{N}(da | \mu_{\theta_\mu}(x), \Sigma_{\theta_\sigma}(x)) \right] \quad (169)$$

$$= \xi \mathbb{E}_{x \sim d_\pi} \left[b_\pi(x)^\top \nabla_\theta \frac{1}{\sqrt{2\pi \Sigma_{\theta_\sigma}(x)}} \int_{\mathcal{A}} k_{\eta_{\pi_{\theta_\sigma}}}(da, \mu_{\theta_\mu}(x)) \right] \quad (170)$$

From (170), the subscript of the kernel indicates the bandwidth of the Gaussian kernel defined as $k_\eta(x, y) = e^{-\eta\|x-y\|^2}$. Moreover, $\eta_\pi = \frac{1}{2\Sigma_{\theta_\sigma}(x)}$. At this point, we leverage Eq. (F.1) in [1], and $\eta_3 = \frac{\eta_{\pi_{\theta_\sigma}} \eta_a}{\eta_{\pi_{\theta_\sigma}} + \eta_a}$ and $a_3 = \frac{\eta_{\pi_{\theta_\sigma}} \mu(x) + \eta_a a_i}{\eta_{\pi_{\theta_\sigma}} + \eta_a}$. Then,

$$\nabla_\theta f = \xi \mathbb{E}_{x \sim d_\pi} \left[b_\pi(x)^\top \nabla_\theta \frac{1}{\sqrt{2\pi\Sigma_{\theta_\sigma}(x)}} \int_{\mathcal{A}} k_{\eta_{\pi_{\theta_\sigma}} + \eta_a}(a, a_3) k_{\eta_3}(a_i, \mu(x)) da \right] \quad (171)$$

Now, we have to derive the gradient with respect to both θ_σ , to update the Σ , and to θ_μ , to update the μ . Generally, the continuous actions are bounded in an interval $[a_{min}, a_{max}]$ and what we should solve is:

$$\nabla_\theta f = \xi \mathbb{E}_{x \sim d_\pi} \left[b_\pi(x)^\top \nabla_\theta \frac{1}{\sqrt{2\pi\Sigma_{\theta_\sigma}(x)}} e^{-\eta_3(a_i - \mu(x))^2} \int_{a_{min}}^{a_{max}} e^{-(\eta_{\pi_{\theta_\sigma}} + \eta_a)(a - a_3)^2} da \right] \quad (172)$$

For simplicity, $[a_{min}, a_{max}]$ is $[a_M, a_m]$

2.1 Gradient wrt θ_σ

Here, we compute the gradient wrt θ_σ . We assume a monodimensional action, and Σ is a single value $\Sigma = \sigma^2$. The computations below:

$$\nabla_{\theta_\sigma} f = \xi \mathbb{E}_{x \sim d_\pi} \left[b_\pi(x)^\top \nabla_{\theta_\sigma} \frac{1}{\sqrt{2\pi\Sigma_{\theta_\sigma}(x)}} e^{-\frac{\eta_{\pi_{\theta_\sigma}} \eta_a}{\eta_{\pi_{\theta_\sigma}} + \eta_a} (a_i - \mu(x))^2} \int_{a_m}^{a_M} e^{-(\eta_{\pi_{\theta_\sigma}} + \eta_a)(a - \frac{\eta_{\pi_{\theta_\sigma}} \mu(x) + \eta_a a_i}{\eta_{\pi_{\theta_\sigma}} + \eta_a})^2} da \right] \quad (173)$$

$$= \xi \mathbb{E}_{x \sim d_\pi} [b_\pi(x)^\top \nabla_{\theta_\sigma} A(x) B_i(x) C_i(x) C_i] \quad (174)$$

We set $M_i(x) = A(x) B_i(x) C_i(x) = A B_i C_i$, where:

$$M(x) = \begin{bmatrix} M_1(x) \\ \vdots \\ M_n(x) \end{bmatrix}. \quad (175)$$

The gradient is:

$$\nabla_{\theta_\sigma} M_i(x) = \frac{\partial M_i(x)}{\partial \Sigma} \nabla_{\theta_\sigma} \Sigma(x) \quad (176)$$

We set $\partial M_i(x) = \partial_\Sigma M_i(x)$ know that:

$$\partial_\Sigma M_i(x) = \partial_\Sigma A B_i C_i + A \partial_\Sigma B_i C_i + A B_i \partial_\Sigma C_i \quad (177)$$

$$= A A^{-1} \partial_\Sigma A B_i C_i + A B_i B_i^{-1} \partial_\Sigma B_i C_i + A B_i C_i C_i^{-1} \partial_\Sigma C_i \quad (178)$$

$$= A B_i C_i A^{-1} \partial_\Sigma A + A B_i C_i B_i^{-1} \partial_\Sigma B_i + A B_i C_i C_i^{-1} \partial_\Sigma C_i \quad (179)$$

$$= M_i(x) [\partial_\Sigma \log A + \partial_\Sigma \log B_i + \partial_\Sigma \log C_i] \quad (180)$$

We compute the gradient of each logarithm below. We remove all the x and θ_σ dependencies from now on to simplify the notation.

$$\partial_\Sigma \log A = \partial_\Sigma \log \frac{1}{\sqrt{2\pi\Sigma}} = -\frac{1}{2\Sigma} \quad (181)$$

$$\partial_\Sigma \log B_i = \frac{\partial \log B_i}{\partial \eta_3} \frac{\partial \eta_3}{\partial \Sigma} \quad (182)$$

$$\partial_{\eta_3} \log B_i = \partial_{\eta_3} \log e^{-\frac{\eta_{\pi\theta_\sigma}\eta_a}{\eta_{\pi\theta_\sigma}+\eta_a}(a_i-\mu)^2} \quad (183)$$

$$= \partial_{\eta_3} \left(-\frac{\eta_{\pi\theta_\sigma}\eta_a}{\eta_{\pi\theta_\sigma}+\eta_a}(a_i-\mu)^2 \right) = \partial_{\eta_3} \left(-\frac{\eta_a}{2\Sigma\eta_a+1}(a_i-\mu)^2 \right) \quad (184)$$

$$= \frac{2\eta_a^2}{(2\Sigma\eta_a+1)^2}(a_i-\mu)^2 \quad (185)$$

we need

$$C_i(\Sigma) = \int_{a_m}^{a_M} e^{-\alpha(a-a_3)^2} \quad (186)$$

where

$$\alpha = \eta_\pi + \eta_a, \quad a_3 = \frac{\eta_\pi\mu + \eta_aa_i}{\eta_\pi + \eta_a}. \quad (187)$$

Both α and a_3 depend on Σ . Therefore,

$$\partial_\Sigma C_i = \int_{a_m}^{a_M} e^{-\alpha(a-a_3)^2} [-(\partial_\Sigma\alpha)(a-a_3)^2 + 2\alpha(a-a_3)\partial_\Sigma a_3] \quad (188)$$

Now compute

$$\partial_\Sigma\alpha = -\frac{1}{2\Sigma^2}. \quad (189)$$

$$\frac{\partial a_3}{\partial \eta_\pi} = \frac{\mu(\eta_\pi + \eta_a) - (\eta_\pi\mu + \eta_aa_i)}{(\eta_\pi + \eta_a)^2} = \frac{\eta_a(\mu - a_i)}{(\eta_\pi + \eta_a)^2} \quad (190)$$

$$\partial_\Sigma\eta_\pi = -\frac{1}{2\Sigma^2} \quad (191)$$

$$\partial_\Sigma a_3 = \frac{\eta_a(a_i - \mu)}{2\Sigma^2(\eta_\pi + \eta_a)^2} \quad (192)$$

$$\partial_\Sigma C_i = \int_{a_m}^{a_M} e^{-\alpha(a-a_3)^2} \left[\frac{1}{2\Sigma^2}(a-a_3)^2 + 2(a-a_3)\frac{\eta_a(a_i - \mu)}{2\Sigma^2(\eta_\pi + \eta_a)} \right] da \quad (193)$$

$$= \frac{1}{2\Sigma^2} \int_{a_m}^{a_M} e^{-\alpha(a-a_3)^2} (a-a_3)^2 da + \frac{\eta_a(a_i - \mu)}{\Sigma^2(\eta_\pi + \eta_a)} \int_{a_m}^{a_M} e^{-\alpha(a-a_3)^2} (a-a_3) da. \quad (194)$$

Define

$$u = a - a_3, \quad du = da.$$

Then the bounds become

$$u_m = a_m - a_3, \quad u_M = a_M - a_3.$$

Therefore,

$$\partial_\Sigma C_i = \frac{1}{2\Sigma^2} \int_{u_m}^{u_M} u^2 e^{-\alpha u^2} du + \frac{\eta_a(a_i - \mu)}{\Sigma^2(\eta_\pi + \eta_a)} \int_{u_m}^{u_M} u e^{-\alpha u^2} du \quad (195)$$

$$= \frac{1}{2\Sigma^2} \int_{a_m-a_3}^{a_M-a_3} u^2 e^{-\alpha u^2} du + \frac{\eta_a(a_i - \mu)}{\Sigma^2(\eta_\pi + \eta_a)} \int_{a_m-a_3}^{a_M-a_3} u e^{-\alpha u^2} du. \quad (196)$$

To solve the integrals we use the following canonical solutions:

Recall that for $\alpha > 0$,

$$\int_{u_1}^{u_2} e^{-\alpha u^2} du = \frac{\sqrt{\pi}}{2\sqrt{\alpha}} [\operatorname{erf}(\sqrt{\alpha} u)]_{u_1}^{u_2}$$

$$= \frac{\sqrt{\pi}}{2\sqrt{\alpha}} (\operatorname{erf}(\sqrt{\alpha} u_2) - \operatorname{erf}(\sqrt{\alpha} u_1)),$$

$$\int_{u_1}^{u_2} u e^{-\alpha u^2} du = \left[-\frac{1}{2\alpha} e^{-\alpha u^2} \right]_{u_1}^{u_2} = -\frac{1}{2\alpha} (e^{-\alpha u_2^2} - e^{-\alpha u_1^2}),$$

$$\int_{u_1}^{u_2} u^2 e^{-\alpha u^2} du = \left[\frac{\sqrt{\pi}}{4\alpha^{3/2}} \operatorname{erf}(\sqrt{\alpha} u) - \frac{u}{2\alpha} e^{-\alpha u^2} \right]_{u_1}^{u_2}.$$

Then, for $\partial_\Sigma \log C_i = \frac{\partial_\Sigma C_i}{C_i}$ and we obtain:

$$\frac{\partial_\Sigma C_i}{C_i} = \frac{\frac{1}{2\Sigma^2} \int_{u_m}^{u_M} u^2 e^{-\alpha u^2} du + \frac{\eta_a(a_i - \mu)}{\Sigma^2(\eta_\pi + \eta_a)} \int_{u_m}^{u_M} u e^{-\alpha u^2} du}{\int_{u_m}^{u_M} e^{-\alpha u^2} du} \quad (197)$$

$$= \frac{\frac{1}{2\Sigma^2} \left[\frac{\sqrt{\pi}}{4\alpha^{3/2}} \operatorname{erf}(\sqrt{\alpha} u) - \frac{u}{2\alpha} e^{-\alpha u^2} \right]_{u_m}^{u_M} - \frac{\eta_a(a_i - \mu)}{2\alpha\Sigma^2(\eta_\pi + \eta_a)} \left[e^{-\alpha u^2} \right]_{u_m}^{u_M}}{\frac{\sqrt{\pi}}{2\sqrt{\alpha}} [\operatorname{erf}(\sqrt{\alpha} u)]_{u_m}^{u_M}} \quad (198)$$

$$\nabla_{\theta_\sigma} f = \xi_{x \sim d_\pi} \mathbb{E} [b_\pi(x)^\top \nabla_{\theta_\sigma} M_i(x)] \quad (199)$$

$$= \xi_{x \sim d_\pi} \mathbb{E} \left[b_\pi(x)^\top M_i(x) \left(-\frac{1}{2\Sigma} + \partial_\Sigma \log B_i + \frac{\partial_\Sigma C_i}{C_i} \right) \nabla_{\theta_\sigma} \Sigma \right] \quad (200)$$

$$= \xi_{x \sim d_\pi} \mathbb{E} \left[b_\pi(x)^\top M_i(x) \left(-\frac{1}{2\Sigma} + \frac{2\eta_a^2}{(2\Sigma\eta_a + 1)^2} (a_i - \mu)^2 + \frac{\frac{1}{2\Sigma^2} \left[\frac{\sqrt{\pi}}{4\alpha^{3/2}} \operatorname{erf}(\sqrt{\alpha} u) - \frac{u}{2\alpha} e^{-\alpha u^2} \right]_{u_m}^{u_M} - \frac{\eta_a(a_i - \mu)}{2\alpha\Sigma^2(\eta_\pi + \eta_a)} \left[e^{-\alpha u^2} \right]_{u_m}^{u_M}}{\frac{\sqrt{\pi}}{2\sqrt{\alpha}} [\operatorname{erf}(\sqrt{\alpha} u)]_{u_m}^{u_M}} \right) \nabla_{\theta_\sigma} \Sigma \right], \quad (201)$$

$$(202)$$

2.2 Gradient wrt θ_μ

Here, we compute the gradient wrt to the mean parameters θ_μ . For this reason, we will omit all θ_σ .

$$\nabla_{\theta_\mu} f = \xi_{x \sim d_\pi} \mathbb{E} \left[b_\pi(x)^\top \nabla_{\theta_\mu} \frac{1}{\sqrt{2\pi\Sigma(x)}} e^{-\eta_3(a_i - \mu_{\theta_\mu}(x))^2} \int_{a_{min}}^{a_{max}} e^{-(\eta_\pi + \eta_a)(a - a_{3\theta_\mu})^2} da \right] \quad (203)$$

$$= \xi_{x \sim d_\pi} \mathbb{E} \left[b_\pi(x)^\top \frac{1}{\sqrt{2\pi\Sigma(x)}} \nabla_{\theta_\mu} e^{-\eta_3(a_i - \mu_{\theta_\mu}(x))^2} \int_{a_{min}}^{a_{max}} e^{-(\eta_\pi + \eta_a)(a - a_{3\theta_\mu})^2} da \right] \quad (204)$$

$$= \xi_{x \sim d_\pi} \mathbb{E} \left[b_\pi(x)^\top \frac{1}{\sqrt{2\pi\Sigma(x)}} \nabla_{\theta_\mu} A_i B_i \right] \quad (205)$$

Then,

$$\partial_\mu A_i B_i = (\partial_\mu A_i) B_i + A_i (\partial_\mu B_i) \quad (206)$$

We split the computations below:

$$\partial_\mu A_i = \partial_\mu e^{-\eta_3(a_i - \mu_{\theta_\mu}(x))^2} \quad (207)$$

$$= 2\eta_3(a_i - \mu_{\theta_\mu}(x)) e^{-\eta_3(a_i - \mu_{\theta_\mu}(x))^2} \quad (208)$$

$$\partial_\mu B_i = \partial_\mu \int_{a_{min}}^{a_{max}} e^{-(\eta_\pi + \eta_a)(a - a_{3\theta_\mu})^2} da \quad (209)$$

$$= \int_{a_{min}}^{a_{max}} \partial_\mu e^{-\alpha(a - a_{3\theta_\mu})^2} da \quad (210)$$

$$= \int_{a_{min}}^{a_{max}} e^{-\alpha(a - a_{3\theta_\mu})^2} \partial_\mu (-\alpha(a - a_3)^2) da \quad (211)$$

$$= \int_{a_{min}}^{a_{max}} e^{-\alpha(a - a_{3\theta_\mu})^2} 2\alpha(a - a_3) \partial_\mu a_3 \quad (212)$$

$$= \int_{a_{min}}^{a_{max}} e^{-\alpha(a - a_{3\theta_\mu})^2} 2\alpha(a - a_3) \frac{\eta_\pi}{\alpha} \quad (213)$$

$$= 2\eta_\pi \int_{a_{min}}^{a_{max}} (a - a_3) e^{-\alpha(a - a_{3\theta_\mu})^2} \quad (214)$$

$$= 2\eta_\pi \left[e^{-\alpha u^2} \right]_{u_m}^{u_M} \quad (215)$$

$$(216)$$

where

$$u = a - a_3, \quad du = da. \\ u_m = a_m - a_3, \quad u_M = a_M - a_3.$$

$$\partial_\mu A_i B_i = (\partial_\mu A_i) B_i + A_i (\partial_\mu B_i) \quad (217)$$

$$= 2\eta_3(a_i - \mu_{\theta_\mu}(x)) e^{-\eta_3(a_i - \mu_{\theta_\mu}(x))^2} \frac{\sqrt{\pi}}{2\sqrt{\alpha}} [\text{erf}(\sqrt{\alpha}u)]_{u_m}^{u_M} + e^{-\eta_3(a_i - \mu_{\theta_\mu}(x))^2} 2\eta_\pi \left[e^{-\alpha u^2} \right]_{u_m}^{u_M} \quad (218)$$

Finally,

$$\nabla_{\theta_\mu} f = \xi \mathbb{E}_{x \sim d_\pi} \left[b_\pi(x)^\top \frac{1}{\sqrt{2\pi\Sigma(x)}} \nabla_{\theta_\mu} A_i B_i \right] \quad (219)$$

$$= \xi \mathbb{E}_{x \sim d_\pi} \left[b_\pi(x)^\top \frac{1}{\sqrt{2\pi\Sigma(x)}} \left(2\eta_3(a_i - \mu_{\theta_\mu}(x)) e^{-\eta_3(a_i - \mu_{\theta_\mu}(x))^2} \frac{\sqrt{\pi}}{2\sqrt{\alpha}} [\text{erf}(\sqrt{\alpha}u)]_{u_m}^{u_M} \right. \right. \quad (220)$$

$$\left. + e^{-\eta_3(a_i - \mu_{\theta_\mu}(x))^2} 2\eta_\pi \left[e^{-\alpha u^2} \right]_{u_m}^{u_M} \right) \nabla_{\theta_\mu} \mu(x) \right] \quad (221)$$

NOTES.

- Questa parametrizzazione sembra essere promettente. Tuttavia, nystrom è necessario
- Usare closed form gradient aiuta efficienza, ma si può usare loss e backprop per autograd, ma sempre usando nystrom.

2.3 The M matrix

When we consider continuous action spaces, the main problem arises in the computation of the M matrix. Indeed, with discrete actions, the integral of the M matrix becomes a summatory and thanks to the Dirac kernel, the computation simplifies a lot. Instead, in the continuous case, the integral must be computed, and smart strategies for encodings and/or kernel must be considered to avoid MC estimation. Here, the point-wise M matrix:

$$(M)_{ij} = \langle \phi(x'_i), P_\pi^* \psi(x_j, a_j) \rangle_{\mathcal{H}} = \langle P_\pi \phi(x'_i), \psi(x_j, a_j) \rangle_{\mathcal{H}} \quad (222)$$

$$= \int_{\mathcal{A}} \langle \psi(x'_i, a), \psi(x_j, a_j) \rangle \mathcal{N}(a; \mu(x_i), \Sigma(x_i)) da \quad (223)$$

$$= \int_{\mathcal{A}} \langle \psi(x'_i, a), \psi(x_j, a_j) \rangle \frac{1}{\sqrt{2\pi}|\Sigma_i|^{\frac{1}{2}}} e^{-\frac{1}{2\Sigma_i}(a - \mu(x_i))^2} da. \quad (224)$$

We consider continuous actions $\mathcal{A} = \mathbb{R}^p$ and assume that the policy is a Gaussian with diagonal covariance, as is common in continuous-control reinforcement learning:

$$\pi(a \mid x_i) = \mathcal{N}(a; \mu_i, \Sigma_i), \quad \mu_i = \mu(x_i) \in \mathbb{R}^p, \quad \Sigma_i = \text{diag}(\sigma_{i,1}^2, \dots, \sigma_{i,p}^2).$$

Option 1: a separable feature map $\psi_\theta(x, a) = W_\theta(x) \otimes \phi(a)$.

For the separable feature map

$$\psi_\theta(x, a) = W_\theta(x) \otimes \phi(a),$$

with $\langle \phi(a), \phi(a') \rangle = k_A(a, a')$, we have that $\langle W_\theta(x) \otimes \phi(a), W_\theta(x') \otimes \phi(a') \rangle = k_X(x, x') k_A(a, a')$ corresponding matrix entry becomes

$$(M_\pi)_{ij} = \langle W_\theta(x_i), W_\theta(x_j) \rangle g_{ij}.$$

We also assume that the action kernel is an unnormalised Gaussian/RBF kernel with diagonal bandwidth matrix

$$k_A(a, a_j) = \exp \left(-\frac{1}{2} (a - a_j)^\top \Lambda^{-1} (a - a_j) \right), \quad \Lambda = \text{diag}(\ell_1^2, \dots, \ell_p^2).$$

Under these assumptions,

$$(a - a_j)^\top \Lambda^{-1} (a - a_j) = \sum_{r=1}^p \frac{(a_r - a_{j,r})^2}{\ell_r^2},$$

and

$$\mathcal{N}(a; \mu_i, \Sigma_i) = \prod_{r=1}^p \frac{1}{\sqrt{2\pi}\sigma_{i,r}} \exp \left(-\frac{1}{2} \frac{(a_r - \mu_{i,r})^2}{\sigma_{i,r}^2} \right).$$

Therefore the action integral appearing in the matrix M_π is

$$\begin{aligned} g_{ij} &:= \int_{\mathbb{R}^p} k_A(a, a_j) \mathcal{N}(a; \mu_i, \Sigma_i) da \\ &= \int_{\mathbb{R}^p} \exp \left(-\frac{1}{2} \sum_{r=1}^p \frac{(a_r - a_{j,r})^2}{\ell_r^2} \right) \prod_{r=1}^p \frac{1}{\sqrt{2\pi}\sigma_{i,r}} \exp \left(-\frac{1}{2} \frac{(a_r - \mu_{i,r})^2}{\sigma_{i,r}^2} \right) da. \end{aligned}$$

Since both the RBF kernel and the Gaussian policy factorise over action dimensions, the integral factorises as

$$g_{ij} = \prod_{r=1}^p g_{ij}^{(r)},$$

where

$$g_{ij}^{(r)} = \int_{\mathbb{R}} \exp \left(-\frac{1}{2} \frac{(a - a_{j,r})^2}{\ell_r^2} \right) \frac{1}{\sqrt{2\pi}\sigma_{i,r}} \exp \left(-\frac{1}{2} \frac{(a - \mu_{i,r})^2}{\sigma_{i,r}^2} \right) da.$$

For a fixed dimension r , define, to simplify notation,

$$\ell := \ell_r, \quad \sigma := \sigma_{i,r}, \quad \mu := \mu_{i,r}, \quad b := a_{j,r}.$$

Then

$$g_{ij}^{(r)} = \frac{1}{\sqrt{2\pi}\sigma} \int_{\mathbb{R}} \exp \left[-\frac{1}{2} \left(\frac{(a - b)^2}{\ell^2} + \frac{(a - \mu)^2}{\sigma^2} \right) \right] da.$$

Expanding the quadratic terms gives

$$\frac{(a - b)^2}{\ell^2} + \frac{(a - \mu)^2}{\sigma^2} = Aa^2 - 2Ba + C,$$

where

$$A = \frac{1}{\ell^2} + \frac{1}{\sigma^2}, \quad B = \frac{b}{\ell^2} + \frac{\mu}{\sigma^2}, \quad C = \frac{b^2}{\ell^2} + \frac{\mu^2}{\sigma^2}.$$

Completing the square,

$$Aa^2 - 2Ba + C = A \left(a - \frac{B}{A} \right)^2 + C - \frac{B^2}{A}.$$

A direct simplification yields

$$C - \frac{B^2}{A} = \frac{(\mu - b)^2}{\ell^2 + \sigma^2}.$$

Hence

$$g_{ij}^{(r)} = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \frac{(\mu - b)^2}{\ell^2 + \sigma^2}\right) \int_{\mathbb{R}} \exp\left[-\frac{1}{2} A \left(a - \frac{B}{A}\right)^2\right] da.$$

Using the Gaussian integral

$$\int_{\mathbb{R}} \exp\left[-\frac{1}{2} A (a - m)^2\right] da = \sqrt{\frac{2\pi}{A}},$$

we obtain

$$g_{ij}^{(r)} = \frac{1}{\sqrt{2\pi}\sigma} \sqrt{\frac{2\pi}{A}} \exp\left(-\frac{1}{2} \frac{(\mu - b)^2}{\ell^2 + \sigma^2}\right).$$

Since

$$A = \frac{1}{\ell^2} + \frac{1}{\sigma^2} = \frac{\ell^2 + \sigma^2}{\ell^2 \sigma^2},$$

we have

$$\frac{1}{\sigma\sqrt{A}} = \frac{\ell}{\sqrt{\ell^2 + \sigma^2}}.$$

Therefore,

$$g_{ij}^{(r)} = \frac{\ell_r}{\sqrt{\ell_r^2 + \sigma_{i,r}^2}} \exp\left(-\frac{1}{2} \frac{(\mu_{i,r} - a_{j,r})^2}{\ell_r^2 + \sigma_{i,r}^2}\right).$$

Finally, multiplying over all action dimensions,

$$g_{ij} = \prod_{r=1}^p \frac{\ell_r}{\sqrt{\ell_r^2 + \sigma_{i,r}^2}} \exp\left(-\frac{1}{2} \frac{(\mu_{i,r} - a_{j,r})^2}{\ell_r^2 + \sigma_{i,r}^2}\right)$$

or, equivalently,

$$g_{ij} = \left[\prod_{r=1}^p \frac{\ell_r}{\sqrt{\ell_r^2 + \sigma_{i,r}^2}} \right] \exp\left(-\frac{1}{2} \sum_{r=1}^p \frac{(\mu_{i,r} - a_{j,r})^2}{\ell_r^2 + \sigma_{i,r}^2}\right).$$

Option 2: a concatenated feature map $\psi_{\theta}(x, a) = \begin{bmatrix} W_{\theta}(x) \\ \phi(a) \end{bmatrix}$.

Suppose

$$\psi_{\theta}(x, a) = \begin{bmatrix} W_{\theta}(x) \\ \phi(a) \end{bmatrix}.$$

Then

$$\langle \psi_{\theta}(x_i, a), \psi_{\theta}(x_j, a_j) \rangle = \langle W_{\theta}(x_i), W_{\theta}(x_j) \rangle + \langle \phi(a), \phi(a_j) \rangle.$$

If

$$\langle \phi(a), \phi(a_j) \rangle = k_A(a, a_j),$$

then

$$(M_{\pi})_{ij} = \int_{\mathbb{R}^p} [\langle W_{\theta}(x_i), W_{\theta}(x_j) \rangle + k_A(a, a_j)] \pi(a | x_i) da.$$

Since the first term does not depend on a,

$$(M_{\pi})_{ij} = \langle W_{\theta}(x_i), W_{\theta}(x_j) \rangle \int_{\mathbb{R}^p} \pi(a | x_i) da + \int_{\mathbb{R}^p} k_A(a, a_j) \pi(a | x_i) da.$$

Because $\pi(\cdot | x_i)$ is a probability density,

$$\int_{\mathbb{R}^p} \pi(a | x_i) da = 1.$$

Therefore,

$$(M_{\pi})_{ij} = \langle W_{\theta}(x_i), W_{\theta}(x_j) \rangle + g_{ij}$$

References

- [1] Alessandro Rudi and Carlo Ciliberto. Psd representations for effective probability models, 2021.

3 Sink state v2

The KME gives a map $\varphi : \mathcal{M}(\mathcal{X}) \rightarrow \mathcal{H}$ from the space of measures to an RKHS. The transfer operator acts like

$$\mathsf{T}\psi(\mu_t) \approx \varphi(\nu_{t+1}). \quad (225)$$

(When T is learned from data we only observe one transition at the time, meaning we only have one data point per each $\mu_t \rightarrow \nu_{t+1}$, but this is irrelevant now).

Our training dataset is made of embeddings $\Psi \sim (n, dA)$, and we can evaluate how a new embedded point $\psi(x, a)$ is represented in the dataset by computing $\|P\psi(x, a)\|$, where P is the projector onto the range of Ψ^* , that is $P = \Psi^\dagger \Psi \approx \Psi^*(\lambda + \Psi\Psi^*)^{-1}\Psi$

4 Virtual state

First, as we agreed to work on measure spaces in 0, let's properly define what embeddings are. To me, $\varphi : \mathcal{X} \rightarrow \partial^d$ so it maps states to measures on a latent, *finite* space. [TODO] A more formally correct definition of embedding is $\varphi : \mathcal{M}(\mathcal{X}) \rightarrow \partial^d$, with domain on the measures on $\mathcal{X}$. My first definition corresponds to the second one restricted to Dirac ∂ 's measures. Through topological arguments, I think we can prove that there exists a reasonable extension of the former to the whole space of measures. Why finite spaces?

1. Uniform distributions are undefined on (non-compact) continuous spaces like $\mathcal{X}$.
2. The whole concept of (differential) entropy, and hence KL divergences, is pathological in continuous space (entropy can be negative).
3. It has a (nice, imo) interpretation as a Hidden Markov Process, where the true dynamics can be simplified into a finite set of states (games fit this intuition rather well).

As in Sec. 1, we start from the problem of matching the state occupancy measure to a target distribution:

$$\pi = \operatorname{argmin}_{\pi \in \Pi} D(\nu_\pi, \nu^*),$$

where

$$\nu_\pi = (1 - \gamma) [\operatorname{Id} - \gamma \mathsf{TP}_\pi]^{-1} \nu_0.$$

When the transfer operator T is learned via least squares

$$\ell(\mathsf{T}; \mathcal{D}) = \sum_{(x, a, y) \in \mathcal{D}} \|\phi(y) - \mathsf{T}\phi(x, a)\|^2 + \lambda \|\mathsf{T}\|,$$

the Thikonov regularization can be interpreted as pushing T towards the zero operator, $\|\mathsf{T} - \mathbb{0}\|$, which maps every state-action distribution into the null element in the space of measures. In particular, if some state x_{unk} hasn't been observed yet, then (x_{unk}, a) will not be represented in the dataset $\mathcal{D}$ for any action, and $\hat{\mathsf{T}}\mathsf{P}_\pi\phi(x) = 0$ and for any policy π Is this *exactly true*?

Question 1.. Now we want to do distribution matching with an estimation of $\hat{\mathsf{T}}$ sending [WIP] What happens to the gradient when $\hat{\mathsf{T}}\mathsf{P}_\pi\phi(x) = 0$? The gradient of the "forward KL" is given by

$$\frac{\nu_0(x)}{(\operatorname{Id} - \hat{\mathsf{T}}\mathsf{P}_\pi)\nu^*(x)} \propto - \frac{|\mathcal{X}|}{1 - \gamma} \frac{\nu_0(x)}{(\operatorname{Id} - \gamma \hat{\mathsf{T}}\mathsf{P}_\pi)\phi(x)} \cdot \nabla_{\mathsf{P}}(\operatorname{Id} - \gamma \hat{\mathsf{T}}\mathsf{P}_\pi) \quad (226)$$

where the target distribution was set to the uniform $\nu^*(x) = |\mathcal{X}|^{-1}$. Up next... I'd like to add a virtual state x^* to the state space, and to change the Tikonov reg term on the LS problem so that it pushes the estimator towards the operator $\phi(x^*) \otimes$

5 Second Eigenvalue Minimization

Here, we define the policy learning objective as the minimization of the second eigenvalue of the matrix M_π . This means that the goal is to reach as soon as possible a stationary state distribution, which, at this point, is not clear in which distribution we end up. Using policy-mirror descent, we have guarantees

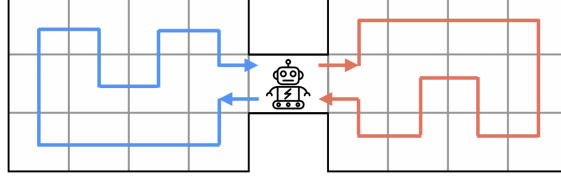


Figure 4: Illustrative two-rooms domain. The agent starts in the middle, colored traces represent optimal strategies to explore the left and the right room.

(?) to not find trivial solutions that are the border of the Δ_{P_π} (i.e., policy simplex), practically speaking, deterministic policies stuck in one of the room corners. However, we are not sure that all the trivial (or sub-optimal) policies are outside the interior of that simplex. Indeed, we can't say anything about the convexity of the problem (?). Given these considerations, let's define the occupancy state-distribution measure and its eigenvalue decomposition as follows.

Diagonalizable case. We assume that the operator M_π is diagonalizable, that is, there exists a complete set of left and right eigenvectors $\{l_i, r_i\}_d$ satisfying $M_\pi r_d = \lambda_d r_d$ and $l_d^\top M_\pi = \lambda_d l_d^\top$, with biorthogonality $\langle l_d, r_{d'} \rangle = \delta_{dd'}$ and $l_i^\top r_i = \text{Id}$. Under this assumption, the operator admits the spectral decomposition

$$M_\pi = R \Lambda L^\top = \sum_d \lambda_d r_d l_d^\top, \quad (227)$$

which allows one to express powers and resolvents of M_π as $(M_\pi)^\top = \sum_d \lambda_d^\top r_d l_d^\top$ and $(\text{Id} - \gamma M_\pi)^{-1} = \sum_d (1 - \gamma \lambda_d)^{-1} r_d l_d^\top$. This representation greatly simplifies the analysis, as each eigenmode evolves independently. We define the discounted state occupancy distribution as

MPPr: Non ho capito perché e se è corretto

$$d_{\nu_0}(\pi) = (1 - \gamma)(\text{Id} - \gamma M_\pi)^{-1} \nu_0 \quad (228)$$

$$= (1 - \gamma) \sum_d (1 - \gamma \lambda_d)^{-1} r_d \langle \nu_0, l_d \rangle \quad (229)$$

$$(M_\pi)^\top \nu_0 = \sum_d e^{-(\frac{1}{\tau_d} + i\omega)t} r_d \langle \nu_0, l_d \rangle \quad (230)$$

5.1 Minimizing Mixing Time by Minimizing SLEM

Let's define the *Second Largest Eigenvalue Modules* $\text{SLEM}(P_\pi) = \max_{i \geq 2} |\lambda_i(P_\pi)|$. We want to minimize the SLEM, to obtain a P_π able to reduce its mixing time. The mixing time of a Markov chain is the time until the Markov chain is "close" to its steady state distribution. This result comes from graph theory, and we should add some references.

Then, we could define our policy mirror descent objective as:

$$P_{\pi_{\pi_t+1}} = \underset{P_\pi}{\text{argmin}} \langle \nabla_{P_\pi} \text{SLEM}(P_{\pi_t}), P_\pi - P_{\pi_t} \rangle + \frac{1}{\eta} D_{\text{KL}}(P_\pi; P_{\pi_t}). \quad (231)$$

As aforementioned, this encourages the policy operator to reach its steady distribution, but we can't guarantee the convergence to any particular distribution without any additional constraint.

Computing the SLEM gradient. In this paragraph, we explicitly compute the gradient $\text{SLEM}(P_{\pi_t})$ in the case M_π is diagonalizable.

$$\lambda_2 = l_2^\top M_\pi r_2 \quad (232)$$

$$\delta\lambda_2 = l_2^\top (\delta M_\pi) r_2 \quad (233)$$

$$M_\pi = T P_\pi \quad (234)$$

$$\delta M_\pi = T \delta P_\pi \quad (235)$$

$$\delta\lambda_2 = l_2^\top (T \delta P_\pi) r_2 \quad (236)$$

$$= \text{Tr}(l_2^\top (T \delta P_\pi) r_2) \quad (237)$$

$$= \text{Tr}(r_2 l_2^\top T \delta P_\pi) \quad (238)$$

$$= \text{Tr}((T^\top l_2 r_2^\top)^\top \delta P_\pi) \quad (239)$$

$$\nabla_{P_{\pi_t}} \lambda_2 = T^\top l_2 r_2^\top \quad (240)$$

$$\nabla_{P_{\pi_t}} \text{SLEM}(P_{\pi_t}) = \nabla_{P_{\pi_t}} |\lambda_2| = \text{sign}(\lambda_2) (T^\top l_2 r_2^\top) \quad (241)$$

What happens when lambda not reals? What if not diagonalizable?

Massi's Idea - don't know the property's name.

$$M_\pi = \sum_d \mu_d \xi_d \psi_d^\top \quad (242)$$

$$\hat{M}_\pi = M_\pi - \mu_0 \xi_0 \psi_0^\top \quad (243)$$

6 Meta RL

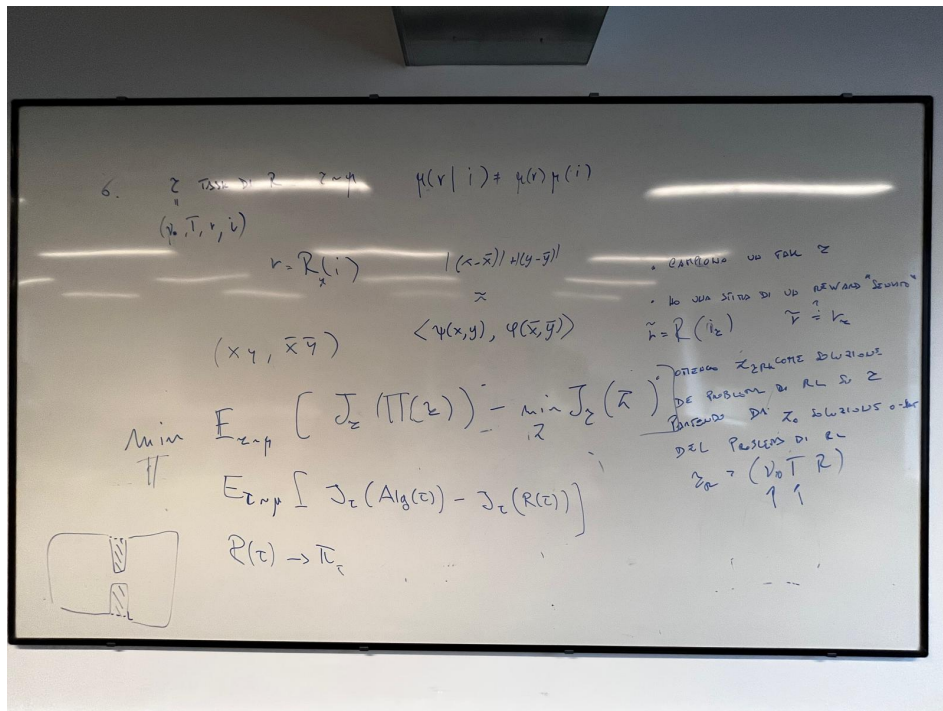


Figure 5: Da scrivere

A Functional Gradients

A.1 Kullback-Leiber

The definition of the KL-divergence is the following:

$$D_{\text{KL}}(\mathcal{P}; \mathcal{Q}) = \sum_{x \in \mathcal{X}} \mathcal{P}(x) \log \frac{\mathcal{P}(x)}{\mathcal{Q}(x)} = \int \mathcal{P}(x) \log \frac{\mathcal{P}(x)}{\mathcal{Q}(x)} dx \quad (244)$$

In the following, we compute the functional derivative with respect to $\mathcal{P}(x)$ or $\mathcal{Q}(x)$.

A.1.1 w.r.t. $\mathcal{P}(x)$

Let's suppose that $\mathcal{Q}(x) > 0$ on the support of $\mathcal{P}(x)$, then:

$$D_{\text{KL}}(\mathcal{P}; \mathcal{Q}) = \int \mathcal{P}(x) \log \frac{\mathcal{P}(x)}{\mathcal{Q}(x)} dx = \int \mathcal{P}(x) \log \mathcal{P}(x) dx - \int \mathcal{P}(x) \log \mathcal{Q}(x) dx \quad (245)$$

$$\delta \frac{D_{\text{KL}}(\mathcal{P}; \mathcal{Q})}{\mathcal{P}(x)} = \frac{\delta}{\delta \epsilon} D_{\text{KL}}(\mathcal{P} + \epsilon \phi; \mathcal{Q}) \Big|_{\epsilon=0} \quad (246)$$

$$= \frac{\delta}{\delta \epsilon} \left[\int (\mathcal{P}(x) + \epsilon \phi(x)) \log(\mathcal{P}(x) + \epsilon \phi(x)) dx - \int (\mathcal{P}(x) + \epsilon \phi(x)) \log \mathcal{Q}(x) dx \right]_{\epsilon=0} \quad (247)$$

$$= \left[\int \phi(x) \log(\mathcal{P}(x) + \epsilon \phi(x)) + \phi(x) dx - \int \phi(x) \log \mathcal{Q}(x) dx \right]_{\epsilon=0} \quad (248)$$

$$= \int \phi(x) (\log \mathcal{P}(x) + 1) dx - \int \phi(x) \log \mathcal{Q}(x) dx \quad (249)$$

$$= 1 + \log \frac{\mathcal{P}(x)}{\mathcal{Q}(x)} \quad (250)$$

A.1.2 w.r.t. $\mathcal{Q}(x)$

$$D_{\text{KL}}(\mathcal{P}; \mathcal{Q}) = \int \mathcal{P}(x) \log \frac{\mathcal{P}(x)}{\mathcal{Q}(x)} dx = \int \mathcal{P}(x) \log \mathcal{P}(x) dx - \int \mathcal{P}(x) \log \mathcal{Q}(x) dx \quad (251)$$

$$\delta \frac{D_{\text{KL}}(\mathcal{P}; \mathcal{Q})}{\mathcal{Q}(x)} = \frac{\delta}{\delta \epsilon} D_{\text{KL}}(\mathcal{P}; \mathcal{Q} + \epsilon \phi) \Big|_{\epsilon=0} \quad (252)$$

$$= \frac{\delta}{\mathcal{Q}(x)} \left[\int \mathcal{P}(x) \log \mathcal{P}(x) dx - \int \mathcal{P}(x) \log(\mathcal{Q}(x) + \epsilon \phi(x)) dx \right]_{\epsilon=0} \quad (253)$$

$$= - \int \frac{\mathcal{P}(x)}{\mathcal{Q} + \epsilon \phi(x)} \phi(x) dx \Big|_{\epsilon=0} \quad (254)$$

$$= - \frac{\mathcal{P}(x)}{\mathcal{Q}(x)} \quad (255)$$

B Temporary derivative of Occupancy Measure Non-discounted

$$\mathcal{O}(s) = \sum_{t=0}^H (\mathbf{TP}_\pi)^t \nu_0(s) \quad (256)$$

$$= \sum_{t=0}^H (M_\pi)^t \nu_0(s) \quad (257)$$

$$\delta M_\pi^t = \sum_{k=0}^{t-1} M_\pi^k \delta M_\pi M_\pi^{t-k-1} \quad (258)$$

$$\delta \mathcal{O}(s) = \sum_{t=0}^H \sum_{k=0}^{t-1} M_\pi^k \delta M_\pi M_\pi^{t-k-1} \nu_0(s) \quad (259)$$

$$= \sum_{t=0}^H \sum_{k=0}^{t-1} M_\pi^k \mathbf{T} \delta \mathbf{P}_\pi M_\pi^{t-k-1} \nu_0(s) \quad (260)$$

$$\delta D_{\text{KL}}(\mathcal{O}; \nu^*) = \left(1 + \log \frac{\mathcal{O}(x)}{\nu^*(x)} \right)^\top \sum_{t=0}^H \sum_{k=0}^{t-1} M_\pi^k \mathbf{T} \delta \mathbf{P}_\pi M_\pi^{t-k-1} \nu_0(s) \quad (261)$$

$$= \sum_{t=0}^H \sum_{k=0}^{t-1} \left(1 + \log \frac{\mathcal{O}(x)}{\nu^*(x)} \right)^\top M_\pi^k \mathbf{T} \delta \mathbf{P}_\pi M_\pi^{t-k-1} \nu_0(s) \quad (262)$$

$$= \sum_{t=0}^H \sum_{k=0}^{t-1} \text{Tr} \left[\left(1 + \log \frac{\mathcal{O}(x)}{\nu^*(x)} \right)^\top M_\pi^k \mathbf{T} \delta \mathbf{P}_\pi M_\pi^{t-k-1} \nu_0(s) \right] \quad (263)$$

$$= \sum_{t=0}^H \sum_{k=0}^{t-1} \text{Tr} \left[M_\pi^{t-k-1} \nu_0(s) \left(1 + \log \frac{\mathcal{O}(x)}{\nu^*(x)} \right)^\top M_\pi^k \mathbf{T} \delta \mathbf{P}_\pi \right] \quad (264)$$

$$= \sum_{t=0}^H \sum_{k=0}^{t-1} \text{Tr} \left[\left((M_\pi^k \mathbf{T})^\top \left(1 + \log \frac{\mathcal{O}(x)}{\nu^*(x)} \right) (M_\pi^{t-k-1} \nu_0(s))^\top \right)^\top \delta \mathbf{P}_\pi \right] \quad (265)$$

$$\nabla_{\mathbf{P}_\pi} D_{\text{KL}}(\mathcal{O}; \nu^*) = \sum_{t=0}^H \sum_{k=0}^{t-1} (M_\pi^k \mathbf{T})^\top \left(1 + \log \frac{\mathcal{O}(x)}{\nu^*(x)} \right) (M_\pi^{t-k-1} \nu_0(s))^\top \quad (266)$$